



**Verified Carbon
Standard**

DELAGUA IMPROVED COOKSTOVE GROUPED PROJECT



Document Prepared by Earthood Services Private Limited

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Summary:

Provide a brief summary of the following:

- A description of the project

The project activity “DelAgua Improved Cookstove Grouped Project” involves distribution of fuel efficient improved cookstoves (ICS) to households in Rwanda. The project is a voluntary initiative as host country laws does not mandate the distribution of ICS to the households. The project proponent is DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) (“DelAgua”) who is using carbon finance to support local partners engaged in different activities like production, distribution, and maintenance of various product technologies. The proposed project activity includes technologies, designed to reduce GHG emission by supplying high efficiency ICS to end-users and promoting public health to tackle several health challenges caused due to using less efficient stoves. This will also reduce the consumption of non-renewable biomass resources or fossil fuel for cooking purposes. The project will also improve the indoor air quality and health of women and children as co-benefits.

- A description of the validation and verification

Earthood Services Private Limited (Earthood) has performed a joint validation & verification of the grouped project activity “DelAgua Improved Cookstove Grouped Project” by “DelAgua Trans Africa Stoves Pte. Ltd. (DATAS)” in Rwanda. This assignment was performed based on rules and requirements defined by VCS for the grouped project activities.

The review of the joint PD & MR, supporting documentation and subsequent follow-up actions (onsite visit and interviews with end-users) have provided Earthood with sufficient evidence to determine the fulfilment of stated criteria. The project correctly applies the approved baseline and monitoring methodology VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1/5/.

The grouped project activity results in reductions of CO₂e emissions that are real, measurable and give long-term benefits to the mitigation of climate change. It is demonstrated that the project is not a likely baseline scenario and the emission reductions attributable to the project are, hence, additional to any that would occur in the absence of the proposed VCS project activity.

The average annual and total GHG emission reduction from the proposed grouped project activity is expected to be 2,149,016 and 17,192,130 tCO₂e, respectively, over the first 7-year crediting period. The monitoring report identified for the first verification period is from 04/11/2022 to 31/12/2022, and the achieved emission reductions during this monitoring period is 59,473 tCO₂e.

- The purpose and scope of validation and verification

The scope of the services provided by the Earthood Services Private Limited is to perform joint validation & verification of the grouped project activity “DelAgua Improved Cookstove Grouped Project”. The scope is to assess the claims and assumptions made in joint project description & monitoring report /1/ against the VCS criteria, including but not limited to the VCS Standard /3/, VCS Validation and Verification Manual /4/ and other relevant rules and requirements established for Verra.

- The method and criteria used for validation and verification

This assignment is an independent and objective review for determination of the monitored reductions in GHG emissions. The joint validation & verification includes the implementation and operation of the project as reported in the project description & monitoring report/1/. The validation & verification process is undertaken by VVB that involves the following:

- The desk review of documents and evidence submitted by the project participant in context of the VCS criteria.
 - Undertaking physical site visit, interviews, or interactions with the representative of the project participant.
 - Reporting audit findings with respect to clarification and non-conformities and the closure of the findings as appropriate.
 - Preparing a draft joint validation & verification report of VCS program complying with the VCS requirements
 - Technical review of the draft validation & verification opinion along with other documents as appropriate by an independent competent technical review team for finalization of the joint validation & verification opinion (this report)
 - An independent technical review team reviews this report made by the assessment team. After the final report is accepted by the Technical Reviewer it is then approved by Earthood Services Private Limited which is processed further according to the VCS procedures.
- The number of findings raised during validation and verification

During joint validation and verification of the project, a total of 03 CLs, 1 CAR and 1 FAR were raised. These have been successfully resolved as provided in Appendix IV of this report.

- Any uncertainties associated with the validation and verification

VVB has undertaken an elaborate audit trail for validation of assumptions and claims contained in final PD & MR Version 8.0, dated 04/11/2024. Each detail has been checked during onsite visit and document review which is not only confined to information provided by PP but also by undertaking an independent evaluation of relevant publicly available information by making use of technical expertise of the assessment team. Thus, there are no evident uncertainties associated with the validation.

- Summary of the validation and verification conclusions

The review of the project description & monitoring report, supporting documentation and subsequent follow up actions have provided ESPL with sufficient evidence to determine the fulfilment of stated criteria. Earthood is of the opinion that the project activity “DeLaGua Improved Cookstove Grouped Project” (VCS ID: 3699) meets all the VCS requirements and has correctly applied the VCS approved monitoring methodology VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1 /5/. The GHG emission reductions were calculated correctly based on the approved methodology.

Earthood Services Private Limited can certify that the emission reductions achieved in the project description & monitoring report for the monitoring period 04/11/2022 to 31/12/2022 by VCS project activity “DeLaGua Improved Cookstove Grouped Project” amount to 59,473 tCO₂e. Therefore, this is being submitted for request for issuance, as per VCS procedures.

In summary, it is Earthood’s opinion that the grouped project activity “DeLaGua Improved Cookstove Grouped Project” as described in the PD & MR, version 8.0, dated 04/11/2024 meets all the relevant baseline and monitoring methodology. The monitoring plan adequately provides for the ex-post monitoring of the project’s emission reductions. The monitoring arrangements described in the monitoring plan are feasible within the project design and it is Earthood’s opinion that the DeLaGua Trans Africa Stoves Pte. Ltd. (DATAS) shall be able to implement the monitoring plan and requirements of VCS standard.

Therefore, Earthood requests the registration of the project as a VCS grouped project activity and issuance of VCUs under the first monitoring period.

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1 INTRODUCTION

1.1 Objective

DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) (PP) has contracted Earthood Services Private Limited (VVB) for joint validation & verification services for the VCS project “DelAgua Improved Cookstove Grouped Project” (project ID 3699) located in Rwanda to jointly carry out the joint validation and verification of the grouped project activity against the requirements of VCS Program.

The objective of the joint validation and verification assessment is to determine the-

- The project complies with all the applicable VCS requirements of the current version of the VCS Standard v4.4/3/ and program guide v4.3/2/;
- Project conformity with the applied methodology, VMR0006: Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1/5/, including the procedure for the demonstration of additionality specified in the applied methodology;
- Likelihood that methods and procedures set out in the project description will generate verifiable GHG data and information when implemented; and
- The data/information reported is accurate, complete, consistent, transparent and free of material error or omission.

In order to fulfil above objectives, the VVB assigned the task of assessment to validation team that is collectively qualified as per Annex A.2.3.3 of ISO 14064-3 and other internal requirements.

The validation/ verification is an independent assessment of the project by a validation/verification body that determines whether the project complies with the VCS rules.

1.2 Scope and Criteria

The joint validation and verification scope of this assignment is defined as an independent and objective review of the VCS Project Description & Monitoring Report (VCS PD & MR) and supporting annexures, which is reviewed by the assessment team against the relevant criteria and decisions by the VCS Board, including but not limited to, the approved baseline and monitoring methodology, applicable tools, VCS standards and guidance. The assessment team has, based on the recommendations prescribed in the latest version of VCS Validation and Verification Manual, employed a risk-based approach, focusing on the identification of significant risks for project implementation and the generation of VCU.

The scope of this assignment is to:

- Jointly conduct validation and verification in accordance with Earthood's own QMS that is based on the VCS VVM along with the guidance provided by the VCS board to determine if the project meets all applicable VCS requirements, including those specified in the VCS standard, relevant methodology, tools, and guidelines and processing the same with VCS Program Guide.
- Asses the accuracy, conservativeness, relevance, completeness, consistency, and transparency of the information provided by the project proponents.
- Determine whether information provided by the project proponents is reliable and credible.
- Present information in the form of joint validation & verification report in a factual, neutral, coherent manner and referencing the sources and assumptions, other forms of validation means employed.
- Report the findings and conclusions in an objective manner and conduct all validation & verification in accordance with VCS rules and procedures.
- Apply consistent validation and verification criteria in providing expert judgments to the requirements of applicable approved methodology, tools, and also cross check the same.
- Safeguard the confidentiality of all information obtained during validation and verification; and
- Adhere to the principles of independence, ethical conduct, fair presentation, and due professional care in assessment process.

1.3 Reasonableness of Assumptions and Level of Assurance

☐ Limited level of assurance

☒ Reasonable level of assurance

A draft joint validation and verification report that is prepared by assessment team is reviewed by an independent technical review team (one or more members) to confirm if the internal procedures established and implemented by Earthood are duly complied with and such opinion/conclusion is reached in an objective manner that complies with the applicable VCS requirements as appropriate. The technical review team is collectively required to possess the technical expertise of all the technical area/sectoral scope the project activity relates to. All team members of technical review team are independent of the validation and verification team. The report approved by Technical Manager is endorsed by Managing Director, who is overall responsible to ensure quality, before final release. The further details of applicable procedures and responsibilities about Earthood Quality Management System (QMS) are available on its website (www.earthood.in).

Earthood's validation approach is based on the understanding of the risks associated with reporting of GHG emission data and the controls in place to mitigate these. Earthood planned and performed the validation by obtaining evidence and other information and explanations that Earthood considers necessary to give reasonable level of assurance that reported estimated GHG emission reductions are fairly stated.

In our opinion the estimated and actual GHG emissions reductions were calculated correctly on the basis of the approved baseline and monitoring methodology VMRO006 Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1 /5/.

The assessment result has a reasonable level of assurance in validation & verification that GHG assertions are free of material errors, omissions, and misrepresentations. The documents and evidence reviewed are included under Appendix I of this report.

1.4 Summary Description of the Project

Summary description of grouped project activity has been illustrated in section 1.1 of Joint PD&MR appropriately. As per the given details in PD, information was assessed on-site and found to be consistent. The project activity “DelAgua Improved Cookstove Grouped Project” involves distribution of fuel efficient improved cookstoves (ICS) to households in Rwanda. The proposed grouped project activity is a voluntary initiative as host country laws do not mandate the distribution of ICS to the households. The project proponent is DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) (“DelAgua”) who is using carbon finance to support local partners engaged in different activities like production, distribution, and maintenance of various product technologies. This project activity includes technologies designed to reduce GHG emission by supplying 500,000 high efficiency ICS to end-users and promoting public health to tackle several health challenges caused due to using less efficient stoves. Under the proposed grouped project, two types of single pot portable stoves will be distributed, which are SSM S32 X stove and Burn Kuniokoa Generation 2. Each household will receive 1 ICS.

The baseline and project scenarios are well described and consistent with the data that was verified during the VVB onsite visit. The average annual and total GHG emission reduction from the proposed group project activity is expected to be 2,149,016 and 17,192,130 tCO₂e, respectively, over the first 7-year crediting period. The monitoring report identified for the first verification period is from 04/11/2022 to 31/12/2022, and the achieved emission reductions during this monitoring period are 59,473 tCO₂e.

2 VALIDATION AND VERIFICATION PROCESS

2.1 Method and Criteria

The validation and verification of the project consists of following steps:

Desk review-

- Review of data and information provided in the VCS joint PD & MR/1/ and corresponding annexures

- Cross check the information with the sources without limitations to the information provided by the project proponent

Follow up actions-

- On-site inspection to physically inspect the project design including, but not limited to, monitoring mechanism
- Interview with relevant personnel and end-users responsible for information given in the VCS PD &MR

Reporting of findings-

- Resolution of findings
- Draft joint validation & verification report

Independent technical review-

The project documentation is reviewed by internal technical reviewer in order to independently confirm whether the applicable GHG program requirements were objectively met or not, in addition to whether internal procedures were followed while arriving at the validation and verification opinion. The technical reviewer may accept or reject the validation and verification opinion prepared by the assessment team and gives the reasons. The resolved findings may be opened at this stage or new findings may be identified that are required to be addressed by assessment team and/or project proponents, as appropriate. The technical reviewer is the decision maker on behalf of the Earthood Services Private Limited. A positive opinion is issued if all the findings have been satisfactorily resolved and in all other cases a negative opinion is issued unless the contract is terminated by either party before reaching the final opinion.

Earthood keeps all documents and records in a secure and retrievable manner for at least two years after the end of the project crediting period.

2.2 Document Review

This joint validation & verification of the grouped project activity is performed primarily as a document review of the VCS Joint PD & MR /1/ and associated documents as stated in detail in appendix I of this document. The assessment is performed by a validation & verification team using a protocol. The cross checks between information provided in the VCS Joint PD & MR and information from sources other than those used, if available, the team's sectoral or local expertise and, wherever necessary, independent background investigations.

The joint validation and verification are performed primarily as a document review of the documents submitted at various stages of assessments. The review is performed by assessment team using dedicated protocols. The assessment team cross checks the information provided in the documents (Joint Project Description & Monitoring Report/1/) and information from sources other than those used,

if available, and also conducts independent background investigations. Earthood conducted a desk review as under;

- a. A review of the data and information presented to verify their completeness.
- b. A review of the monitoring plan, the monitoring methodology including applicable tool(s) and, where applicable, the applied standardized baseline, paying particular attention to the frequency of measurements, the quality of project technology including efficiency of the project stove, and the quality assurance and quality control procedures
- c. An evaluation of data management and the quality assurance and quality control system in the context of their influence on the generation and reporting of emission reductions

2.3 Interviews

The VVB conducted an on-site visit from 19/01/2023 to 20/01/2023. The on-site visit was conducted as a part of validating the project and also verifying the monitoring procedures followed in the project. The following points were discussed as the part of the interviews:

- a. Introduction, scope and objective of work, roles and responsibilities of audit team, resources required, and timetable of the onsite audit including venue for closing meeting and any concerns from PP.
- b. All the details about the project activity mentioned in the Joint PD-MR was cross checked by discussion with the PP.
- c. Compliance of monitoring procedures followed at project site with registered PDD and monitoring methodology.
- d. Conducting the Site visit by interviewing the end-users.
- e. And finally, submission of the audit findings to the client and agreement on the issues raised and agreement on timelines.

The list of PP representatives interviewed by the VVB team is as follows:

Name	Affiliation	Date	Subject	Assessment team members
Ajay Dixit	Carbon Operations Manager (Delagua)	19/01/2023 to 20/01/2023	- Project design - Project Implementation status - Project start date and Project Location	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Monica Keza	Country Director			
Robert Masinde	Supply Chain Manager			
Jean Pierre Sibomana	Operation Manager			
Bruno Nyirimana	Program Coordinator			
Maria Chistelle Ishimwe	Program Coordinator			

Francis Regis Ruzirabwoba	Program Coordinator		- Choice of applied methodology - Baseline Identification and Additionality of project - Grouped project eligibility criteria - Monitoring and reporting procedure - Quality Assurance - Management and operating systems - local stakeholder consultation process - Compliance with relevant laws	
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The list of LSC participants remotely interviewed through the Microsoft Teams application by the VVB are as follows:

Name	Affiliation	Date	Subject	Assessment team members
Didane Mishimwe	Rwanda Development Board	04/05/2023	- Date of LSC Conducted	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Annonciata	Rwanda Environment Journalists Association		- Feedback about the project	
Maniragaba Augustin	Project ICS user		- Continuous Grievance Mechanism	
Nyirabera Domithile	Project ICS user			

2.4 Site Visits

The VVB carried out an on-site visit from 19/01/2023 to 20/01/2023 and physically inspected the project technical design and implementation as specified in the VCS PD & MR /1/.

The site visit was conducted to validate the accuracy and completeness of the project description. The validation and verification team members who visited the site collectively had the competence for the relevant technical area to which project relates. The views obtained during site observation were considered while concluding the validation and verification opinion.

The sampling plan adopted by the VVB for carrying out the joint validation and verification assessment is as follows;

- **Sampling plan for validation of the project:**

The Sampling and surveys for CDM project activities and programmes of activities (Version 9.0)/7/ states under paragraph 28 that “When the project participants or the coordinating/managing entity have applied a sampling approach, the VVB may apply acceptance sampling as described in the steps indicated in paragraphs 29–38 below as part of validation/verification activities”.

Since the PP has determined the value of ex-ante parameter B_{old} , baseline stove type, fuel type through sampling of baseline end-users, the validation team conducted acceptance sampling in line with paragraph 30 and 31 of the sampling standard version 9.0/7/.

As per para 39 of the sampling standard version 9.0/7/, “A VVB may select a different sample size than the one indicated in paragraph 32 above either by choosing a different value for the consumer risk and producer risk (e.g., 20 per cent for the consumer risk) when applying acceptance sampling or by using another approach, if any of the following conditions apply:

(a) The estimated volume of annual GHG emission reductions of the project activity or the PoA being verified is equal to or less than 100,000 tCO₂e.

(b) The security conditions in the project region prevents inspection of many samples (eg. conflict zones);
or

(c) The VCS grouped project activity is located in a least developed country or a host party with 10 or fewer registered CDM project activities at the end of the monitoring period being verified.”

The grouped project is located in Rwanda, which is a Least Developed Country (LDC) (<https://unfccc.int/topics/resilience/workstreams/national-adaptation-programmes-of-action/ldc-country-information>). Thus, the validation team can choose a different sample size than one indicated in para 32 of the sampling standard/7/.

The validation team selected random sample of PP’s baseline survey records (parameter- B_{old}) to check the acceptability (or otherwise) of the data for each such record with PP’s sample records and determined if the PP’s sample records meet the requirements.

Sample Size:

AQL	UQL	Producer Risk	Consumer Risk	Sample Size; Min	Acceptance No.
0.5%	20%	10%	20%	8	0

The Validation team covered a total of 8 samples to confirm baseline KPT results. All the households interviewed were found to have inefficient three stone stove using firewood and confirmed that KPTs were conducted. Therefore, no discrepancies were observed.

- **Sampling plan for verification of the project:**

The CDM Standard – Sampling and surveys for CDM project activities and programme of activities (Version 9.0)/7/ states under paragraph 28 that “When the project participants or the coordinating/managing entity have applied a sampling approach, the VVB may apply acceptance sampling as described in the steps indicated in paragraphs 29–38 below as part of validation/verification activities”.

Since the PP has monitored parameters ($N_{y,i,j}$, η_{old} , μ_y) through surveys, verification team conducted acceptance sampling in line with paragraph 30 and 31 of the sampling standard version 9.0.

As per para 39 of the sampling standard version 9.0/7/, “A VVB may select a different sample size than the one indicated in paragraph 32 above either by choosing a different value for the consumer risk and producer risk (e.g., 20 per cent for the consumer risk) when applying acceptance sampling or by using another approach, if any of the following conditions apply:

(a) The estimated volume of annual GHG emission reductions of the project activity or the PoA being verified is equal to or less than 100,000 tCO₂e.

(b) The security conditions in the project region prevents inspection of many samples (eg. conflict zones); or

(c) The VCS grouped project activity is located in a least developed country or a host party with 10 or fewer registered CDM project activities at the end of the monitoring period being verified.”

The grouped project is located in Rwanda, which is a Least Developed Country (LDC) (<https://unfccc.int/topics/resilience/workstreams/national-adaptation-programmes-of-action/ldc-country-information>). Thus, the verification team can choose a different sample size than one indicated in para 32 of the sampling standard/7/.

The verification team selected random sample of PP’s monitoring survey records (parameter- $N_{y,i,j}$, η_{old} , μ_y) to check the acceptability (or otherwise) of the data for each such record with PP’s sample records and determined if the PP’s sample records meet the requirements.

Sample Size:

AQL	UQL	Producer Risk	Consumer Risk	Sample Size; Min	Acceptance No.
0.5%	20%	10%	20%	8	0

The verification team covered a total of 8 samples to confirm monitoring survey results. All the households interviewed were found to have inefficient three stone stove using firewood and confirmed that the project stove was operational. Therefore, no discrepancies were observed by the verification team.

The list of end-users interviewed by the assessment team is as follows:

Name	Affiliation	Date	Subject	Assessment team members
Uzamukunda Theophile	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Mukarugambwa Jeanne	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Mukankundiye Marie	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Nyirareme Berthilde	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Mbonimpa Clementine	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Uwamahoro Clementine	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Tumukunde Josephine	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Uwimana Anastasia	Baseline KPT sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana

Name	Affiliation	Date	Subject	Assessment team members
Umurerwa Liliane	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Mukanubaha Josephine	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Samuel Rusingizandekwe	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
JMV Mugiraneza	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Mukeshimana Jacqueline	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Bizimana	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
Jean De dieu Nzarubara	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana

Delphine Ahishakiye	Monitoring survey sample	19/01/2023 to 20/01/2023	VVB field survey	Jahnabi Kalita, Deepika Mahala and Olivier Uwimana
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2.5 Resolution of Findings

The validation and verification protocols used for the assessment is designed in accordance with the latest version of latest VCS Program Documents. The validation and verification protocols are an internal document in Excel and available on request.

As an outcome of the joint validation and verification process, the assessment team can raise different types of findings. Clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS/CDM requirements have been met. Where a non-conformance arises the validator/verifier raised a Corrective Action Request (CAR). A CAR is issued, where:

- The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions.
- The CDM/VCS requirements have not been met.
- There is a risk that emission reductions cannot be monitored or calculated.

Forward Action Request (FAR) is raised during validation/verification to highlight issues related to project implementation that require review during the first verification of the project activity. FARs do not relate to the VCS/CDM requirements for registration.

Irrespective of the nature of findings, all of these are given to PP in a separate finding document. In this document, the project proponent(s) is/are given the opportunity to respond to the findings. Based on the responses by PP along with relevant supporting documents/evidence the validation team determines whether the findings are resolved or not. The findings may also be raised at later stage e.g., during internal technical review process and these are also communicated/dealt in the same manner as described above. The draft joint validation & verification report is also prepared during this period reflecting the status of the findings, if any, as appropriate.

In summary, the type and total number of findings that were raised are indicated below;

Type of Finding	CAR	CL	FAR
Total Findings	01	03	01

All the findings raised during this joint validation & verification of the project are included under Appendix IV of this report.

2.5.1 Forward Action Requests

01 Forward Action Request (FAR) was raised during this assessment.

3 VALIDATION FINDINGS

3.1 Project Details

Project type, technologies and measures implemented, and eligibility of the project

The project “DelAgua Improved Cookstove Grouped Project” falls into sectoral scope 03 and is appropriately selected in the VCS PD & MR and is implemented in Rwanda. The project category and type are correctly indicated in the VCS PD & MR/01/. It has been indicated as well as accepted that it is a grouped project activity.

The project entity i.e., DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) is correctly listed in the VCS PD. No other entity has been listed for this grouped project.

The start date of the Grouped project activity is 04/11/2022/9/, which is also the date of installation of first ICS under the grouped project. The start date of the Grouped project activity is identified in accordance VCS standard/3/.

The crediting period of the Grouped project activity is from 04/11/2022 to 03/11/2029 and the estimated emission reductions from the Grouped project activity is 17,192,130tCO₂e from the entire crediting period and the annual emission reduction for the group project activity is expected to be 2,149,016 tCO₂e.

The project crediting period is correctly filled in VCS PD, which is of 7 years and twice renewable, for a maximum total of 21 years. The selected crediting period is consistent with the project start date and the first crediting period spans from 04/11/2022 to 03/11/2029 (both days included).

The project is correctly classified as large project as per VCS Standard /3/ as the average annual emission reductions are more than 300,000 tCO₂e. The estimated annual average emission reductions are estimated as 2,149,016 tCO₂e and the total emission reductions over the crediting period of 7 years are 17,192,130 tCO₂e. The estimated emission reductions are consistent with the corresponding ER spreadsheet /8/.

Project design, including eligibility criteria for grouped projects

The project activity involves distribution of improved cookstoves in households of Rwanda. The project is a voluntary initiative as host country laws do not mandate the distribution of ICS to the local communities. The project proponent is DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) (“DelAgua”) who is using carbon finance to support local partners engaged in different activities like production, distribution, and maintenance of various product technologies. This project includes technologies, designed to reduce GHG emission by supplying high efficiency ICS to end-users and promoting public health to tackle several health challenges caused due to using less efficient stoves. The description of the ICS models is given in PD & MR/1/ and was confirmed from manufacturer’s specification/10/.

The grouped project activity envisages distribution of two models of ICS under the initial PAIs. The details (provided below) of the ICS were confirmed from the manufacturer’s specifications/10/ and third party efficiency reports/37,38,39/:

Model-1:

Name of the stove – Kuniokoa Generation 2 with Pot Skirt

Stove manufacturer – Burn

Stove Type – Natural Draft

Stove Body – Stainless Steel

Stove height – 32.1 cm

Stove diameter – 28.2 cm

Thermal efficiency with pot skirt – 51.3%

Thermal efficiency without pot skirt – 43.4%

Stove lifetime – 10 years

Model-2:

Name of the stove – S32-X With Pot Skirt

Stove manufacturer – SSM

Stove Type – Natural draft

Stove Body – Stainless Steel

Stove height – 28 cm

Stove diameter – 32 cm

Thermal efficiency with Pot Skirt – 50.1%

Thermal efficiency without Pot Skirt – 45.7% Stove lifetime – 10 years

The increase in efficiency of the project stoves can be attributed to the addition of pot skirts. Moreover, Generation 2 stove models have been distributed under this project which are an improved version of the G1 (Generation 1) stove model. The G2 stoves have higher efficiency than the previous version (G1) of the same model. VVB has checked the third-party efficiency reports of ICS (BURN Kuniokoa G2 and SSM S32-X) with and without pot-skirt models and confirmed that the thermal efficiency values used by PP is accurate/ 37,38,39/. Further, during the VVB onsite visit to the households with the distributed project stove (BURN Kuniokoa G2), the use of pot skirt was confirmed by the end-users while cooking and they also confirmed that use of pot skirt was beneficial for reducing cooking time and providing better efficiency in cooking.

The VCS PD & MR/1/ includes sufficient information regarding the project activity, project stoves description and distribution plan. Further description on application and information on the ICS technology is provided under section 1.11 of the PD & MR/1/.

Project eligibility under VCS is assessed below

A. Eligibility conditions

Scope	Applicability	VVB Assessment
The seven Kyoto Protocol greenhouse gases.	The GP reduces GHG emissions associated with the combustion of cooking fuels. Gases included are CO ₂ , CH ₄ , N ₂ O.	PP has considered the GHG gases within the scope of Kyoto protocol which are reduced using improved cookstove under the grouped PA.
Ozone-depleting substances (ODS).	The GP does not involve any ODS.	The proposed grouped project does not involve any Ozone-depleting substances as it involves only dissemination of ICS.
Project activities supported by a methodology approved under the VCS Program through the methodology approval process.	The ICS to be distributed) in the GP uses the methodology “VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1.” which is a VCS approved methodology.	The grouped project applies, VMR0006 “Methodology for Installation of High Efficiency Firewood Cookstoves”, version 1.1/5/, applicability of the methodology is assessed in detail under section 3.4 of this report.
Project activities supported by a methodology approved under a VCS approved GHG program, unless explicitly excluded under the terms of Verra approval.	Not applicable.	Not applicable

B. Excluded project activities under the VCS Program

Excluded Activity	Applicability	VVB Assessment
Grid-connected electricity generation using hydro-power plants/units	N/A The GP reduces GHG emissions associated with the combustion of cooking fuels.	The proposed grouped project disseminates ICS which is not related to hydro-power plants. Thus, exclusion condition is not applicable.
Grid-connected electricity generation using wind, geothermal, or solar photovoltaic (PV) power plants.	N/A. The GP activities are energy efficiency measures and do not generate electricity from wind, geothermal, or photovoltaic (PV) power plants.	The proposed grouped project disseminates ICS which is not related to wind, geothermal, or photovoltaic (PV) power plants. Thus, exclusion condition is not applicable.

Activities recovering waste heat for combined cycle electricity generation, or to heat/cool via cogeneration or trigeneration.	N/A The GP activities are energy efficiency measures and do not utilize recovered waste heat for any purposes.	The project does not involve any component of electricity generation or heating/cooling via cogeneration or trigeneration. Thus, exclusion condition is not applicable.
Activities generating electricity and/or thermal energy for industrial use from the combustion of non-renewable biomass, agro-residue biomass, or forest residue biomass.	N/A. The GP activities are energy efficiency measures and do not generate electricity using biomass. The GP includes energy efficiency improvements in thermal applications, i.e. cookstoves.	The project does not involve any component of electricity generation. Thus, exclusion condition is not applicable.
Activities generating electricity and/or thermal energy using fossil fuels, and activities that involve switching from a higher to a lower carbon content fossil fuel.	N/A. The GP activities do not include the generation of electricity and/or thermal energy using fossil fuels, including activities that involve switching from a higher carbon content fuel to a lower carbon content fuel.	The project does not involve any component of electricity generation. Thus, exclusion condition is not applicable.
Activities replacing electric lighting with more energy-efficient electric lighting, such as the replacement of incandescent electrical bulbs with compact fluorescent lights (CFLs) or light emitting diodes (LEDs).	N/A The GP activities do not include the replacement of electric lighting.	The project does not include any replacement of electric lighting. Thus, exclusion condition is not applicable.
Activities installing and/or replacing electricity transmission lines and/or energy-efficient transformers.	N/A. The GP activities do not include the installation and/or replacement of electricity transmission lines and/or energy efficient transformers.	The project does not involve any component of electricity generation. Thus, exclusion condition is not applicable.

Activities that reduce hydrofluorocarbon-23 (HFC-23) emissions	N/A. The GP reduces GHG emissions associated with the combustion of cooking fuels	The proposed grouped project disseminates ICS which does not involve HFC-23. Thus, exclusion condition is not applicable.
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Eligibility criteria to include PAIs:

S No	Criterion	How the new project activity instances comply	VVB Assessment
1	Project Activity Instances (PAIs) must meet the applicability conditions set out in the applied methodology.	See below	See below
1a	Project activities shall be implemented in domestic premises or in community-based kitchens.	Each PAI will distribute technology that is appropriate for domestic use to household end-users	PP will provide ICS appropriate only for domestic scale. The specification of the ICS to be distributed were checked from Manufacturer's specifications/10/ confirming the target users for future PAIs. Project ICS distribution database/8/ containing unique household and end user information was checked along with physical visit to the households to confirm the target end users of the PAI 1 which is already implemented. The locations coordinates were checked using get-geocoordinates app.
1b	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of	ICS distributed in each PAI will have a thermal efficiency of at least 25% and will be confirmed by the end user that they were using the firewood as the cooking fuel in the project's baseline.	The ICSs distributed till now is Burn Kuniokoa. Other than this S3X stove is planned to be distributed under the other PAIs. The technical description of both the stoves were checked from the Manufacturer's specifications/10/. The efficiency

	project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	Each PAI will record the baseline and project fuel use to ensure that wood fuel is the sole source of energy.	of the ICS were further confirmed from third party lab test report/37,38/. The design efficiency of both the stoves are higher than 25%. Information recorded in the baseline KPT sheet/24/ on the baseline fuel usage were corroborated through VVB survey to the households on which baseline KPTs were performed. During monitoring the fuel used in project scenario will be confirmed.
1c	Both 'Projects' and 'Large Projects' can use the methodology	PAIs will comply with the GP requirements	The GP is defined as a "Large Project" in Joint PD and MR inline with para 3.10.1 of VCS Standard /3/. The scale of the project was also confirmed through review of estimated ER sheet/8/.
1d	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	The PP has demonstrated at grouped project level that Non-renewable biomass has been used in the project region since 31 December 1989 under section 3.2.	The use of Non-renewable biomass since 1989 was confirmed through published report/12,13/.
1e	For the specific case of biomass residues processed as a fuel (e.g., briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored	Biomass residues are not processed as a fuel to be used in the GP. End users will be asked to state their source of energy and baseline appliance at the point of distribution and during the project monitoring. Above information will be stored in the monitoring database.	The project is not conducting any activity to process biomass. The monitoring plan include monitoring surveys to confirm the fuel used in the project scenario. PP had shown the Project ICS distribution poster during the VVB site visit, which explains the end-users to correctly use the ICS with wood fuel. Project ICS distribution database/8/ and baseline survey

	during the crediting period and I Energy use for renewable biomass processing (e.g., shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.		records/24/ were checked to confirm that that fuel used by end users under PA1 are not processed.
1f	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo)	To avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient. Each PAI will have system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 1. Date of distribution 2. A unique serial number on each technology 3. Available end-user data: name, phone number, address/physical location, ID number	The proposed group project will maintain database with end user details and the UID of the device they receive. The device has unique engravings or text specifically mentioning the unit is strictly “not for sale” along with the DelAgua logo to avoid double counting with any other GHG project in the same area. Moreover, the households are checked for baseline stove in use before distribution of the new project ICS in the household. VVB confirms that following data points were provided in the database submitted: 1. Date of distribution 2. Geographic area of distribution 3. Model of the ICS distributed 4. Names of the recipients and actual end users. 5. Phone numbers of end-users wherever available. 6. Address of the end users (District, Sector, Cell, Village) 7. Unique identification ID (alpha numeric) of each

			device The uniqueness of entries could be confirmed by cross-checking for any repetitive entries in the database in various combinations of the data points listed above. Upon further review w.r.t. household name and recipient name, it was found that in 76 entries, the household name and recipient name were identical. To clarify this, PP provided link to the distribution forms with photo IDs confirming that while some individuals share the same household and recipient name, they are distinct individuals with different date of birth and photo. Upon checking these distribution forms, it was found that, in a total of 50 forms, households shared the same name, but they are completely different individuals with distinct date of birth and different photos. For 26 entries where photo IDs were not clear due to poor photo quality, the PP has conservatively removed these 26 samples from the database. Project ICS distribution database/8/ was checked to confirm that the ICSs distributed under PA1 are not part of any other Voluntary GHG programme.
1g	The Project Design Document shall also explain how the proposed procedures prevent double counting of emission	The End User Agreement confirms that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their	End users will be asked to sign end user agreement to confirm that they transfer the rights to GHG on receipt of stoves for new PAIs.

	reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	understanding of this agreement will be collected at the point of distribution. The End User Agreement will be in English and translated into a local language where required. If necessary, it will be read out to the End User. All rights to distribute the delivered stoves and claim the resultant GHG emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.	For the PAI1 included under the grouped PA, sample agreements have been checked to confirm transfer of GHG rights/14/.
2	Use the technologies or measures specified in the project description.	Only ICS that conform with the GP description are to be distributed in the project. The ICS will be chosen to deliver a level of service at least equivalent to the baseline appliance.	The ICSs distributed till now under PAI 1 is Burn Kuniokoa. Other than this S3X stove is planned to be distributed under the other PAIs. The technical description of both the stoves were checked from the Manufacturer's specifications/10/. The design efficiency of both the stoves are higher than 25% as confirmed from the 3rd party lab test report/37,38,39/.
3	Apply the technologies or measures in the same manner as specified in the project description.	The ICS distributed in the PAIs will adhere to the GP description (i.e. burning of biomass for cooking meals) and replace the baseline device which also was utilising the biomass burned in the baseline scenario.	The PAIs will include only the ICS shown to be meeting methodology applicability criterion in the Joint PD and MR/1/. The same has been followed for PAI 1/8/.
4	Are subject to the baseline scenario determined in the project description for the	All new PAIs will be implemented only in regions within the geographic borders of Rwanda	The PP will record the baseline stove at the time of distribution and document in ICS Distribution

	specified project activity and geographic area.	subject to the same baseline scenario determined in Section 3.4 in this Project Description. The geographical location of the project ICS can be traced through the information provided in the distribution database, which includes the latitude and longitude data of the ICS distributed.	database. It was confirmed during the physical interviews with the end users that traditional woodstove has been replaced by the project stove. Project ICS distribution database /8/ was checked to confirm that all the ICS distributed under PA1 was in Rwanda.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area.	All new PAIs will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the applicability conditions of the methodology. Where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional.	An analysis of Rwandan government programmes or policy for cookstoves as shown under Section 3.5 of Joint PD and MR/1/ was carried out that there is no mandate for the distribution of ICS to the households in Rwanda. The PP has provided the End User Agreement to confirm that ICS under PAI 1 were distributed at zero cost. This was also cross checked by the VVB during the interviews with the end-users. Thus, the PAI is considered auto additional. Similar approach will be followed for future PAIs.
6	Where a capacity limit applies to a project activity included in the project, no PAI shall exceed such limit.	The GP does not have any specified capacity limits applicable, since there are no capacity limits defined within the	The applied methodology does not specify capacity limits/5/.

	Further, no single cluster of project activity instances shall exceed the capacity limit, determined as follows: Each PAI that exceeds one percent of the capacity limit shall be identified. Such instances shall be divided into clusters, whereby each cluster is comprised of any system of instances such that each instance is within one kilometre of at least one other instance in the cluster. Instances that are not within one kilometre of any other instance shall not be assigned to clusters. None of the clusters shall exceed the capacity limit and no further project activity instances shall be added to the project that would cause any of the clusters to exceed the capacity limit.	applied methodology, VMR0006, ver. 1.1. Therefore, this eligibility criterion is not applicable.	
Eligibility conditions specific to inclusion of New PAIs			
7	The new PAI must occur within one of the designated geographic areas specified in the project description.	The distribution of the ICS will be done in the jurisdictions of Rwanda. Each PAI distributes the ICS units to HHs in umudugudus (Kinyarwanda term used for villages) and cells which are distinct from each other. No two PAI or VCS Grouped projects in the host country	Project stove will be distributed only in the host country, Rwanda. Project ICS distribution database/8/ will record the household locations. The condition has been set correctly in line VCS standard v4.4/3/.

		Rwanda can have ICS units distributed to same HH. This is ensured by accessing the beneficiary list from the cell office, which is then verified on-site by the DelAgua team for any corrections in the number of beneficiaries. Post that, the final list of beneficiaries is reviewed by the umudugudu (village) chief. Distribution related information of ICS units specific to each PAI and other DelAgua grouped project activities is then recorded in separate databases using electronic data recording software, form.com. The unique identifiers of the distributed ICS unit reflect that it is distributed to a unique HH in the host country Rwanda.	
8	The new PAI must comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	The eligibility criteria set must be complied by the PAIs at all points in time throughout the crediting period. If at any point, it is found that the PAI is not complying with the set of eligibility criteria, then it may be excluded from the crediting for the duration of the monitoring period, until it is able to demonstrate the compliance of the eligibility criteria.	The new PAIs will demonstrate conformation with eligibility criteria in the MR. The condition has been set correctly in line VCS standard v4.4/3/.
9	The new PAI must be included in the monitoring report with sufficient technical, financial, geographic and other	The new PAI will include all the technical, financial, geographic and other relevant information related to the compliance of the applicable set of eligibility criteria	The new PAIs will demonstrate conformation with eligibility criteria in the MR in line with Joint PD and MR. The condition has

	relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	& also enable sampling by the VVB.	been set correctly in line VCS standard v4.4/3/.
10	The new PAI must be validated at the time of verification against the applicable set of eligibility criteria.	The new PAI(s) shall be part of the monitoring, and therefore, will be validated against the criteria specified, by an VCS accredited/approved VVB.	The new PAIs will demonstrate conformation with eligibility criteria in the MR in line with Joint PD and MR. The condition has been set correctly in line VCS standard v4.4/3/.
11	The new PAI must have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	The PAI related evidence on the project ownership will be in the form of a distribution end user receipt poster. This will be signed by the end user, and a photograph of him/her holding the poster along with the ICS will be saved in the PAI database.	The start date of the new PAIs will be determined through the installation documents of the first unit which will be checked at the time of inclusion. The condition has been set correctly in line VCS standard v4.4/3/. The start date of the PA1 is 04/11/2022/9/, which is also the date of installation of first ICS under the grouped project.
12	The new PAI must have a start date that is the same as or later than the grouped project start date.	The new PAI will have a start date later than the GP start date.	The start date of the new PAIs will be determined through the installation documents of the first unit disseminated under them which shall be after the start date of the grouped project. The condition has been set correctly in line VCS standard v4.4/3/. The PAI-1, is first instance and start date of grouped project activity is same as start date pf PAI-1. The start date of the Grouped project activity is 04/11/2022/9/, which is also

			the date of installation of first ICS under the grouped project.
13	The new PAI must be eligible for crediting from the start date of the project activity instance through to the end of the project crediting period.	The PAI will be only eligible for the crediting from the start of the verification period in which they were added to the grouped project.	The start date of the new PAIs will be determined through the installation documents of the first units disseminated under them. The end date will be cross checked with the end date published on VCS registry. The condition has been set correctly in line VCS standard v4.4/3/. The PAI-1, is first instance and start date of grouped project activity is same as start date pf PAI-1, hence no overlapping of crediting period.
14	The new PAI shall not leave one VCS project and subsequently enroll in another VCS project.	It will be ensured that the PAI that is being enrolled under the VCS GP 3699 was not previously enrolled under any other VCS project, and will not be enrolled under other VCS project if it is removed from the VCS GP 3699.	PP will provide a signed declaration ensuring that new PAIs that will be enrolled under the VCS GP 3699 will not be enrolled under another VCS project. The validation team has verified that the PAI1 has not been registered, and it has no plans to register under any other GHG program by reviewing the declaration from the PP/19/.
15	The new PAI shall adhere to clustering requirement that states the project proponent shall include in a singular project all project activity instances within ten kilometers of another instance of the same project activity and with the same project proponent	It will be ensured that PAI included within 10 kilometers of another instance of the same project activity by the same PP is in a singular project	There are no other project instances by the same PP (DeIagua Trans Africa Stoves Pte. Lte. (DATAS)) distributed within 10 kilometres for the same project activity as confirmed from the independent research by the validation team.

CL#02 has been raised and resolved.

Project proponent and other entities involved in the project

DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) is the project proponent of the project. No other entity is involved in this grouped project activity.

Ownership

DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) is the sole owner of the carbon credits generated from the VCS grouped project activity titled “DelAgua Improved Cookstove Grouped Project”. This has been clearly mentioned in the VCS PD & MR section 1.7 /1/ and was confirmed by reviewing manufacturer’s agreement/15/. Also, end-user agreement is signed with each end-user receiving project technology (ICS) to confirm the ownership of rights are only held by PP and end-users have voluntarily waived their rights on carbon credits generating from the use of project stove by signing the poster agreement. VVB confirms that all the evidences demonstrating the ownership of the project or right to use were found to be acceptable. It was found to be in compliance with section 3.7.1 of VCS standard/3/. Independent research was conducted to confirm that the project is not listed under any emission trading program or other binding limits.

The Project proponent for Grouped Project 3699 has been changed from DelAgua Health Rwanda (Voluntary) Limited (DHRV) to DelAgua Trans Africa Stoves Pte. Ltd. (DATAS). DelAgua received interest from an investor for their cookstove program in Rwanda (VCS 3699). As per the initial investment condition, the project (VCS 3699) had to be registered with “DelAgua Trans Africa Stoves Pte. Ltd. (DATAS)” as the project proponent (PP). The company incorporation process along with account opening was taking time. In the meantime, to keep the investor’s confidence, VCS ID 3699 was submitted for the request for registration with DelAgua Health Rwanda (Voluntary) Limited (DHRV) as PP. Considering the project is yet to be registered and all the required legalities for DATAS are already completed prior to project registration, PP has requested VERRA to accept the request for PP change. PP has also submitted Deed of Accession for transferring VCS ownership from DHRV to DATAS.

FAR#01 has been raised,

Project start date

The start date of the grouped project activity, as identified by the PP is 04/11/2022, which is also the date of installation of first ICS under the Grouped project activity/9/. It was found to be in compliance with section 3.8 of VCS standard/3/.

Project crediting period

The project crediting period will be seven years, twice renewable, for a maximum total of 21 years. The first crediting period will be 04/11/2022 to 03/11/2029 (both days included).

Project scale and estimated GHG emission reductions or removals

The scale of the project is large project, and the estimated average annual GHG emission reductions are 2,149,016 tCO₂e. The total emission reductions over the crediting period of 7 years are 17,192,130 tCO₂e. The estimated emission reductions are consistent with the corresponding ER spreadsheet /8/.

Project location

The project location is Rwanda, and all the project activity instances will be implemented within the boundaries of Rwanda. The project boundary for each Project Activity Instance is the physical, geographical location of the ICS distributed within the host country. The geographical coordinates of the project site were verified at the time of monitoring survey samples during VVB on-site visit. Get-geocoordinates mobile application was used to check the geographical coordinates for the samples visited on-site. The information made available by the PP, the KML files shared of the distribution districts within Rwanda/16/ and the geo-coordinates obtained by VVB/17/ have been verified during the site visit and found to be consistently defining the project location as the geographical boundaries of the host country, Rwanda. Therefore, this was found to be acceptable by the VVB.

Conditions prior to project initiation

The baseline scenario of the project consists of use of non-renewable biomass (firewood/charcoal) for cooking in households throughout the project location.

Project compliance with applicable laws, statutes, and other regulatory frameworks

As this grouped project is a distribution of ICS technologies, this does not come under purview of Environment Impact Assessment (EIA) as per Rwandan laws and regulatory framework. This was verified through General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda /18/.

Participation under other GHG programs

○ Projects registered (or seeking registration) under other GHG program(s):

The validation team has verified that the grouped project activity has not been registered, and it has no plans to register under any other GHG program by reviewing the declaration from the PP that the project has not claimed or will not claim carbon credits under any other scheme after registration of the proposes project under VCS/19/.

○ Rejection by other GHG programs:

The project has not been rejected by any other GHG programs/19/.

Other forms of credit and supply chain (Scope 3) emissions

○ Emissions trading programs and other binding limits:

The PP confirms in section 1.16.1 of the joint PD & MR /1/ that this grouped project does not reduce GHG emissions from activities that are included in an emissions trading program. Rwanda does not have its own ETS. Also, it does not seek or receive another form of GHG-related environmental credit.

- **Other forms of environmental credit sought or received and eligible to be sought or received:**

The grouped project activity did not seek or receive another form of GHG-related environmental credit.

- **Issuance of public statement(s) to help prevent Scope 3 emissions double claiming:**

The grouped project activity which aims at distribution of ICS to households does not result in any Scope 3 emissions in its supply chain.

- **Email notification of the potential risk of Scope 3 emissions double claiming:**

The grouped project activity does not result in any Scope 3 emissions, hence not applicable.

Additional information relevant to the project, including

- **Leakage management:**

As the grouped project activity adopts a net gross adjustment factor of 95% to account for leakage, this is not applicable.

- **Commercially sensitive information:**

The VCS PD&MR/1/ does not indicate any information that is commercially sensitive in nature as confirmed by PP under section 1.18 of joint PD&MR/1/.

- **Sustainable development contributions:**

The project activity is contributing to 8 SDGs for the PAI 1:

SDG 1 No Poverty – Indicator 1.4.1 Proportion of population living in households with access to basic services. In the current monitoring period, 100% of PAI-1 users i.e., 130,633 households have access to basic cooking services as confirmed from the monitoring survey when compared with the baseline.

SDG 3 Good Health and Well Being – Indicator 3.9.1 Mortality rate attributed to household and ambient air pollution. In the current monitoring period, 561,834 people have less risk towards respiratory ailments due to smoke inside their homes due to the implementation of the project activity. While 100% of PAI1 users did experience less smoke and pollution after using the project device and this was confirmed from the monitoring survey.

SDG 4 Quality Education – Indicator 4.7.1 Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment. The number of CHWs trained by the end of current monitoring period, are 4,783. 130,633 people were educated in PAI-1 and this has been explained in the PD/MR.

SDG 5 Gender Equality – Indicator 5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location. In the current monitoring period, 97.5% of PAI-1 users did experience less time in collecting fuelwood as confirmed from the monitoring survey. This was multiplied with the total number

of end user HHs, resulting in 127,367 HHs spending less time on unpaid domestic work (i.e. collection of firewood for cooking) during the first monitoring period.

SDG 7 Affordable and Clean Energy – Indicator 7.1.2 Proportion of population with primary reliance on clean fuels and technology. In the current monitoring period, 130,633 households received ICS in PAI-1 and this was confirmed from the distribution database.

SDG 8 Decent Work and Economic Growth – Indicator 8.3.1 Proportion of informal employment in total employment, by sector and sex. In the current monitoring period, the project activity provided informal employment to around 6000 people.

SDG 13 Climate Action – Indicator 13 Take urgent action to combat climate change and its impacts. During the current monitoring period, the GHG emission reductions results to 59,473 tCO₂e and this was confirmed from the ER calculation sheet.

SDG 15 Life on Land – Indicator 15.4.2 Mountain Green Cover Index. In the current monitoring period, 426,568 tons of woody biomass was saved while implementing PAI-1 under its first monitoring period.

All the SDGs listed in the PD/MR have been assessed by the VVB thoroughly by desk review and also cross checked during the on-site visit done by VVB.

PP also mentioned in the PD & MR that this grouped project also seeks certification under SD Vista and the monitoring and reporting of these SDGs will be demonstrated as per the requirements of SD Vista.

No further information is included in the VCS Joint PD & MR/1/. This is to confirm that the information in the project description is accurate, complete, and provides an understanding of the nature of the project.

3.2 Participation under Other GHG Programs

The grouped project activity has not been registered, and it has no plans to register under any other GHG program. The validation team has verified by reviewing the declaration from the PP that the project has not claimed or will not claim carbon credits under any other scheme after registration of the project under VCS/19/. Therefore, the grouped project activity is inline with section 3.22.1 of VCS standard/3/ and is eligible to participate under the VCS Program.

3.3 Safeguards

3.3.1 No Net Harm

As this grouped project is a distribution of ICS technologies, this does not come under purview of Environment Impact Assessment (EIA) as per Rwandan laws and regulatory framework. This was verified through General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda /18/. It was further identified in the PD & MR /1/ that as per as Ministerial Order No. 004/2008: Ministerial Order which establishes the list of works, activities and projects that must undertake an environment impact assessment, this project of distribution of ICS to households in Rwanda does not

require EIA. VVB accepted this justification that since project does not need environmental assessments, supports that project does not cause environmental harm.

3.3.2 Local Stakeholder Consultation

The local stakeholder consultation process has been described, by the PP, in Section 2.2 of the VCS PD&MR/1/. The Project Participant correctly identified the relevant stakeholders such as local NGOs, representatives of local authorities, target project ICS users. Local stakeholder consultation process was conducted through inviting the relevant stakeholders to comment on the project activity. The PP conducted the local stakeholder consultation meeting on 01/03/2023 which was attended by representatives of local NGOs, authorities, and other interested stakeholders. The participants across the host country (Rwanda) were invited and informed about the meeting through email invite, newspaper advertisement and FM Radio announcement notifying the date, time and location of the event /31/. The attendance sheet, photos and videos of the local stakeholder meetings were submitted as evidence by the PP/35,36/.

All questions raised by stakeholders were answered and documented by the PP and their representatives. Stakeholders had no comments/complaints/grievances by the end of the meeting which could have any significant modification in the project description or its design. The LSC modules/34/ for the 3 districts: Gakenke, Gicumbi and Musanze were submitted as feedback and no negative comments were received from any of them. The date and mode of invitation were confirmed from remote interviews of the LSC participants conducted by validation team. VVB validation team was also able to confirm that the remotely interviewed stakeholders were part of the LSC meetings conducted by the PP and were found to be aware about the proceedings of the meeting including benefits of the improved cookstove, including lesser smoke, lesser time in wood collection, lesser wood consumption, etc. As per the VVB's assessment, PP has addressed all questions and responses of all the stakeholders with satisfaction and for future grievances PP has provided the communication details that can be used for placing the complaints after the implementation of the project.

The PP has a continuous grievance mechanism in place for the local stakeholders. The beneficiaries of the project ICS have been provided with the following means to contact the PP:

- Directly registering complains at the nearest cell office.
- A toll-free helpline on which the end users can seek assistance.
- Speaking to the DeAgua CHW network to share their feedbacks.

Evidences on the collection of grievance feedback from the end-users is provided by PP (photographs of the grievance/feedback notes), and the requirement is identified as being fulfilled during the VVB physical visit.

CL#01 has been raised in this context.

3.3.3 Environmental Impact

No environmental impact assessment has been done for this project as the project type is exempted from EIA as per Rwandan laws and regulations. Please see section 3.3.1 of this report for further clarification.

3.3.4 Public Comments

This project was open for public comment from 10/11/2022 to 10/12/2022. No comments were received by the project during the public comment period as confirmed from the project registry webpage /20/.

3.3.5 AFOLU-Specific Safeguards

This project is a non-AFOLU project type, therefore this section is not required.

3.4 Application of Methodology

3.4.1 Title and Reference

The project, as given in the section 3.1 of VCS Joint PD & MR /1/, has applied VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1/5/ and relevant tools prescribed therein. The reference to the versions of the tools and methodology were found to be valid for use. The validity of the versions of the applied methodology and tools were checked from Verra and CDM websites.

Tools applied in this project are listed below:

CDM TOOL01 “Tool for the demonstration and assessment of additionality” Version 07

CDM TOOL21 “Demonstration of additionality of small-scale project activities” Version 13.1

CDM TOOL30 “Calculation of the fraction of non-renewable biomass” Version 3.0

CDM Guideline “Sampling and surveys of CDM project activities and programmes of activities” version 4.0

CDM Standard “Sampling and surveys for CDM project activities and programmes of activities” version 9.0

3.4.2 Applicability

The project “DelAgua Improved Cookstove Grouped Project” complies with each applicability condition of the identified methodology /5/. The same has been justified in the following table.

As per para 3.8.1. of VCS standard/3/, non-AFOLU projects shall complete validation within two years of the project start date. Since the project start date was determined to be 04/11/2022. The validation can be completed within 2 years i.e., by 03/11/2024. It shall be noted that the project applies latest version of the methodology/5/.

Applicability criteria of applied methodology, VMR0006 for PAIs to be included are listed in the table below:

No.	Applicability Condition	Justification of applicability	Means of assessment
1	Project activities shall be implemented in domestic premises or in community-based kitchens.	Since the PAI distributes technology that is appropriate for domestic use to household end users i.e. improved cookstoves for cooking, thus this meets the applicability	The aim of the project is to reduce GHG emission and provide improved cookstoves following this methodology/5/ to households in Rwanda. It was verified from the ICS distribution database/8/ and VVB onsite visit that project stoves were distributed to only household within the project boundary i.e., Rwanda
2	The project stove shall have specified high-power thermal efficiency of at least 25% per the manufacturer's specifications and shall exclusively use woody biomass and can be single pot or multi-pot; in case of project stove replacing fossil fuel baseline stove, it shall exclusively use renewable biomass.	ICS distributed in each PAI has a thermal efficiency of at least 25% Baseline and project fuel for each PAI has been recorded to ensure that wood fuel is the sole source of energy.	Thermal efficiency of all the ICS will be more than 25% which shall be demonstrated at the time of PAI inclusion. The ICSs distributed till now is Burn Kuniokoa. Other than this SSM S32X stove is planned to be distributed under the other PAIs. The design efficiency of both the stoves are higher than 25% as checked from the Manufacturer's specifications/10/ and third party lab test report /37,38,39/. Therefore, the criterion has been met.
3	Both 'Projects' and 'Large Projects' can use the methodology	As per the VCS requirements, the VCS GP 3699 is classified under the Large Project category.	The GP is defined as a "Large Project" in Joint PD and MR inline with para 3.10.1 of VCS Standard /3/. The scale of the project was also confirmed through review of estimated ER sheet/8/.

		As such, PAIs also fall under the Large Project category.	Thus, the grouped project meets the Group project requirements set out in the methodology/5/.
4	Non-renewable biomass has been used in the project region since 31 December 1989, using survey methods or referring to published literature, official reports or statistics.	Statistics from the Forest department show that forests were estimated to cover 659,000 ha in 1960, which has reduced to 240,746 ha in 2007¹. This reflects a loss of approximately 64 per cent of forests between 1960 and 2007, which is more than 1.3 per cent per year. Firewood remains the main source of domestic energy for more than 90 per cent of Rwandan households, s. The above can also be confirmed from Forest Resource Assessment report 2020 for Rwanda, which states around 45.27% loss in forest from 1984 to 2015². Further the study conducted in June 2009 in various districts of Rwanda reveals that 98% of rural households use wood fuels as their main source of energy for cooking, which consists of	The use of Non-renewable biomass since 1989 was confirmed through published report/12,13,32,33/. It will be demonstrated for the future PAIs as well. Therefore, the criterion has been met.

¹ <https://www.rema.gov.rw/soe/chap6.pdf>

² FRA 2020 Étude de bureau, Rwanda

		92% firewood, 5% charcoal and 3% other residue³. A recent study conducted by Ministry of Land and Forestry, Rwanda in 2017⁴, reveals that wood demand and supply ratio is 2:1 and the shortage is projected to increase in the future unless alternative sources of wood energy are sought. . The Business as Usual scenario on wood supply/demand, estimates the deficit between wood supply and demand to be 4.3 million tonnes (oven dry weight) in 2017, which is projected to increase to 7.5million tonnes by 2026. This is due to a high increase demand for fire wood and wood for charcoal. This must imply over-exploitation of already low stocked forests. The analysis also shows that 93% of total wood demand mentioned above is for cooking purpose (57% firewood and 36% charcoal). It can therefore be concluded that non-renewable	
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³ <https://www.cleancookingalliance.org/binary-data/RESOURCE/file/000/000/36-1.pdf>

⁴ https://www.climateinvestmentfunds.org/sites/cif_enc/files/fip_final_rwanda.pdf

		biomass has been used in Rwanda since long back before 31st December 1989.	
5	For the specific case of biomass residues processed as a fuel (e.g. briquettes, wood chips), it shall be demonstrated that: (a) It is produced using exclusively renewable biomass (more than one type of biomass may be used). (b) The consumption of the fuel should be monitored during the crediting period and I Energy use for renewable biomass processing (e.g. shredding and compacting in the case of briquetting) may be considered as equivalent to the upstream emissions associated with the processing of the displaced fossil fuel and hence disregarded.	Biomass residues are not processed as a fuel to be used in the GP. The PAI uses non-renewable biomass (firewood) as source of fuel for the improved cookstove. No processed biomass residue is used.	The project involves distribution of ICS products and does not involve with biomass residues; therefore, this criterion is not applicable.
6	The Project Design Document shall explain the proposed method for distribution of project devices including the method to avoid double counting of emission reductions such as unique identifications of product and end-user locations (e.g., programme logo)	In order to avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient.	Each PAI will maintain a distribution database recording date of distribution and UID of the product as a mechanism to avoid double counting. The unique ID associated with each ICS product distributed to end-users were checked during VVB site visit and were found as correct and consistent with the monitoring survey records/11/ and the distribution database/8/.

		Each PI will have system for avoiding double counting with data being collected via a smartphone app (where possible), or via paper forms, including: 1. Date of distribution 2. A unique serial number on each technology 3. Available end-user data: name, phone number, address/physical location, ID number	
7	The CDM-PDD or CDM-PoA-DD/CPA-DD shall also explain how the proposed procedures prevent double counting of emission reductions, for example to avoid that project stove manufacturers, wholesale providers or others claim credit for emission reductions from the project devices.	The End User Agreement confirms that the ownership rights of the ER/VCUs generated by the PAI reside with the PP. The signature of each end user confirming their understanding of this agreement will be collected at the point of distribution. The End User Agreement is in English and translated into a local language where required. If necessary, it will be read out to the End User. All rights to distribute the delivered stoves and	End user agreement/14/ will be collected from each household to transparently show the transfer of GHG rights. It was cross checked by the VVB during the onsite interviews with the end-users that each household had a signed agreement which confirms that the ownership of the resultant carbon credits belong solely to the PP.

		claim the resultant emissions reductions belong to the PP. The GP will not include any PAI that has been rejected by another GP, or PA-DDs of any other GHG programs.	
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3.4.3 Project Boundary

The project boundary and identified GHG sources, sinks and reservoirs for the project and baseline scenarios are appropriately defined in the VCS PD & MR /1/. All the project activity instances (PAIs) that will be added in within the project boundary i.e., Rwanda. The selection and justification for inclusion or exclusion is appropriate and duly supported in the VCS PD & MR/1/.

The PP has considered CO₂, CH₄ and N₂O as major GHG in the baseline scenario and project scenario. Emission from burning of non-renewable biomass has been considered in both the scenario. The approach was found to be in line with the applied methodology/5/.

3.4.4 Baseline Scenario

The baseline scenario for ICS has been identified using the methodology VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves, version 1.1 /5/. As per section 6 of applied methodology, the baseline scenario is the continued use of non-renewable wood fuel (firewood/charcoal) or fossil fuel (coal/kerosene) by the target population to meet similar thermal energy needs as provided by project cookstoves in absence of project activity.

The project activity involves distribution of improved cookstoves to the households in Rwanda. In the absence of project activity, the cookstoves used in the households are of three stone fire and conventional stoves with less efficient combustion system. As informed in the PD section 1.1/1/, the energy and mass flows, and the type and level of service provided through improved cookstoves in project scenario will remain same as of baseline scenario. This has been ensured by checking manufacturer's specification of ICS introduced through this project /15/ and independent desk review. PP conducted baseline survey and baseline KPTs between September to December 2021, to assess the type & quantity of fuels consumed and the cooking technologies in use in pre-project scenario. The baseline scenario was identified as the continued use of traditional cookstove (e.g. Rondereza) without chimney/ grate or three-stone fired for preparation of meal. More than 90% of households in the rural and semi-urban areas are relying on firewood for cooking. PP used guidance provided in para 8.4-b of VMR0006 version 1.1. to conduct the baseline survey. Sample size was determined using the applied methodology VMR0006 ver 1.1 and KPT protocol. Guidance provided in para 8.4-b of VMR0006 version 1.1. has been used to conduct the baseline survey. A sample size of 150 was considered for Baseline KPT considering the non-response scenario of the survey. According to the methodology applied/5/, since the population size is

more than 1000, the minimum sample size of 100 is required to be covered during the baseline survey, which is achieved in this scenario. PP has randomly selected 32-36 samples per province, totaling 137 households. Baseline KPTs were also conducted on 137 households (accounting for non responses) which is also in line with KPT protocol. .

The procedures for identifying the baseline scenario have been correctly followed and the identified scenario reasonably represents what would have occurred in the absence of the project. VVB confirms that the applied methodology, tools, and documentary evidence/21,22,23/ are correctly quoted and interpreted. VVB has checked the baseline KPT data and baseline survey submitted by PP. The information was also cross-checked with the households during the on-site visit.

The PP has also established the baseline fuel consumption in the host country by using recent study “National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda” conducted by the Ministry of Infrastructure and Ministry of Finance, the Government of Rwanda in the year 2020/40/ which has reported the fuel consumption by households at 6.75 tonnes per year. A comparative analysis for the value of baseline fuel consumption for various registered projects across Rwanda (Verra and Gold Standard) was also conducted. The *B_{old}* value used in the registered projects under Verra (VCS ID: 3654 and 2380) varies between 6.75 to 7.6 tonnes per year/41,42/. Region-specific projects are also registered under Gold Standard which reports the value of baseline fuel between 3.17 to 4.97 tonnes per year. However, these do not represent the entire country of Rwanda and hence, these values cannot be compared for the proposed project. Therefore, the VVB concludes that the value of fuel consumption derived using baseline KPTs (4.4092 tonnes/year) is conservative when comparing the achieved value with other sources.

The assumption and data can be deemed reasonable. The VCS PD & MR/1/ considers the national and sectoral policies in effect. The selected methodology /5/ has been correctly applied to identify the baseline scenario as what would happen in the absence of the project and in compliance with the prescription given therein.

3.4.5 Additionality

PP has demonstrated additionality according to the applied methodology /5/ as given below:

The activity method is used by the PP to demonstrate additionality as prescribed by VMR0006, v1.1 /5/. The project is not mandated by Rwandan laws and the project is deemed additional as the project meet all the applicability conditions of the methodology and also comply with the positive list. Under the additionality section of the applied methodology /5/, it is stated that where the project activity installs or distributes stoves at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits, the project activity shall be deemed additional. Since the project is distributing improved cookstoves at zero cost to the end-user, as cross checked by the VVB during the interviews with the project beneficiaries/17/. Thus, the project is deemed additional.

Therefore, VVB confirms that the project additionality has been sufficiently established proved by the PP in accordance with the applied methodology VMR0006, v1.1 /5/.

3.4.6 Quantification of GHG Emission Reductions and Removals

The project applies the VCS methodology VMR0006, v1.1/5/. All the parameter considered for calculation of emission reductions are correctly applied by the PP.

Data and Parameters fixed ex-ante:

S.No.	Parameter	Value applied	Means of assessment
1.	B _{old}	4.4092 tonnes/year	The parameter represents annual quantity of woody biomass that would have been used in the household in the absence of the project activity to generate useful thermal energy equivalent to that provided by the project devices. According to methodology /5/, the value of B_{old} can be sourced from historical data or baseline surveys. The source of the data has been taken from baseline KPT surveys/24/ and was found correct and acceptable upon checking the KPT excel sheet /24/. The description, source, and justification of choice of data is included in accordance with applied methodology /5/. It is to be noted that all KPT analysis was conducted in a calculation sheet/24/ where raw data collected from site was directly linked with the calculation of average wood consumption through excel formulae. To further explain, there is an initial tab in KPT calculation sheet “Raw data” which consists of all questions and corresponding responses/ records (wood use on each day, moisture meter readings, number of people- male, female with respective age groups, notes and observations, etc.) of visited and surveyed households which were covered as part of the KPTs. The next two tabs “Analysis-Part1” indexes these readings from “Raw data” tab using excel functions and wood used each day is thus calculated. “Analysis -Part2”, a subsequent data tab in the KPT calculations sheet then further refines the data (which is linked to the previous tab) by removing all entries with potential data errors and also removing any outliers followed by calculation of average consumption of fuelwood per day (which is subjected to confidence and precision test). Therefore, raw data is directly used in the calculation, thus leaving no scope of error due to data transfer. The VVB can therefore confirm that data has been consistently reported. To further confirm consistent and accurate reporting of raw wood measurements from survey records to the KPT calculation sheet, the baseline survey sheets generated from phone based application were shared with VVB/52/. 18 samples were picked by VVB using random sample generator/53/ following the principles

		of acceptance sampling and survey sheets for these samples were provided by PP. The following factors were considered for determining the sample size in line with “Sampling and surveys for CDM project activities and programmes of activities” version 9.0: AQL: 1%, UQL: 20%; Producer risk: 5%, Consumer risk: 10% Sample aize-18, acceptance number-1 The values reported in these sheets were compared with the wood use data reported in KPT calculations sheet. The values were found consistently and accurately reported in the ‘Raw data’ tab of KPT analysis excel file, as confirmed on basis of the acceptance sampling. Sample size calculation: A sample size of 150 was considered for Baseline KPT considering the non-response scenario of the survey/24/. According to the methodology applied/5/, since the population size is more than 1000, the minimum sample size of 100 is required to be covered during the baseline survey, which is achieved in this scenario. Sample Selection criteria: Rwanda has a total of 4 provinces – Eastern, Western, Northern, Southern, and the city of Kigali. Since Kigali is predominantly an urban region and since this project primarily targets rural households, the approach to conduct baselines surveys in other four provinces is found to be acceptable. As per literature/50/, cooking behavior in the entire country is quite similar. The households are culturally, and socio-economically homogenous and therefore, random sampling across the country was found to be an acceptable approach. The targeted sample size of 150 were picked using a randomized approach while ensuring representativeness of all four provinces. The actual sample coverage (accounting for non-responses) is as follows: <table><tr><th>Province</th><th>Baseline KPT Samples Covered</th></tr><tr><td>Eastern</td><td>35</td></tr><tr><td>Northern</td><td>34</td></tr><tr><td>Southern</td><td>32</td></tr></table>	Province	Baseline KPT Samples Covered	Eastern	35	Northern	34	Southern	32
Province	Baseline KPT Samples Covered									
Eastern	35									
Northern	34									
Southern	32									

			<table><tr><td>Western</td><td>36</td></tr><tr><td>Total</td><td>137</td></tr></table>	Western	36	Total	137	
Western	36							
Total	137							
			Procedures followed: PP conducted baseline KPTs between September to December 2021, which was aligned with the “lean period” prior to harvesting season in order to ensure conservative estimation of baseline wood consumption without leading to biasness in the baseline KPT results due to seasonal variations of wood harvesting affecting household income or food consumption as outlined in the KPT protocol/47/. The KPTs were conducted for 4 days for baseline scenario fuelwood consumption as verified from KPT excel sheet /24/. The KPT surveys were conducted by a competent team trained for the KPT surveys by Berkeley Air Monitoring Group (BAMG) as confirmed during onsite interviews with PP representatives and cross-checked with training certificate of the staff/51/. A digitized data collection approach was taken by the PP team, where the survey responses and measurement records were recorded in the electronic survey forms through the smartphones. The survey responses for each individual respondent were recorded in the centralized database, where all the survey results were combined, and the final KPT survey excel sheet/24/ was exported. The results of the baseline KPT survey were checked through acceptance sampling to check the acceptability (or otherwise) of PP’s baseline KPT survey records by cross checking physical site visit with the households which were picked up as samples. During the onsite interviews, the validation team directed questions to the end-users about the baseline KPTs namely: • Stove in use (in baseline)? • Fuel used in the baseline stove? • How many days was the KPT conducted in 2021? Which month during the year of 2021? • Did Berkeley Air monitoring group visit your house to conduct KPT in 2021? • Was the household provided with wood (fuel) prior to conducting the KPT? • Did the monitoring group visit daily to measure wood consumption by the household? Were they visiting at the same time of the day?					

			The PP provided the statistical analysis in the file “the KPT excel sheet /24/ which was checked by the assessment team. The aggregated data satisfies the 90/10 rule, i.e., the endpoints of the 90% confidence interval in each case lie within $\pm 10\%$ of the estimated mean. The precision achieved for the baseline KPT survey results was $95\% \pm 6.47/24/$. The value of 4.4092 tonnes/ year was derived from direct measurements of daily firewood consumption using digital scales during the baseline survey. The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/.
2.	η_p	Kuniokoa Generation 2 0.4340 for project stove without skirting 0.513 for project stove with pot skirting SSM S32X: 0.501 for project stove with skirting 0.457 for project stove without skirting	The parameter represents efficiency of project stove at the start of project activity. The value of the parameter is sourced from manufacturer’s specification/10/ and third party lab test reports/37,38,39/. The increase in efficiency of the project stoves can be attributed to the addition of pot skirts. Moreover, Generation 2 stove models have been distributed under this project which are an improved version of the G1 (Generation 1) stove model. The G2 stoves have higher efficiency than the previous version (G1) of the same model. VVB has checked the third-party efficiency reports of ICS (BURN Kuniokoa G2 and SSM S32-X) with and without pot-skirt models and confirmed that the thermal efficiency values used by PP is accurate./ 37,38,39/. Further, during the VVB onsite visit to the households with the distributed project stove (BURN Kuniokoa G2), the use of pot skirt was confirmed by the end-users while cooking and they also confirmed that use of pot skirt was beneficial for reducing cooking time and providing better efficiency in cooking. The value applied was found valid and correct. The description, source, and justification of choice of data is included in accordance with applied methodology /5/. The value is consistent in PD & MR/1/ and ER calculation sheet/8/.
3.	Life Span	Burn Kuniokoa Generation 2: 10 years SSM S32X:	The parameter represents operating lifetime of project device for projects opting Equation 6 $\eta_{new, i,y} = \eta_p \times (DF_n)^{y-1} \times 0.94$ for updating project stove efficiency during project crediting period. The value of the parameter is sourced from manufacturer’s specification/15/. The value applied was found valid and correct.

		10 years	
4.	NCV_{biomass}	0.0156 TJ/tonne	The parameter represents net calorific value of the non-renewable woody biomass. The value has been sourced from IPCC /26/ and was found correct and valid. The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/.
5.	$EF_{\text{wf CO}_2}$	112 tCO ₂ /TJ	The parameter represents CO ₂ emission factor for woody biomass. The value has been sourced from IPCC/26/ and was found correct and valid. The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/ and was determined in line with the applied methodology/5/.
6.	$EF_{\text{wf non-CO}_2}$	26.23 tCO ₂ /TJ	The parameter represents non-CO ₂ emission factor for woody biomass. The value has been sourced from IPCC /26/ and was found correct and valid. The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/ was determined in line with the applied methodology/5/.
7.	LF	0.95	The parameter represents the leakage factor applied to account for increase in NRB use outside the project boundary. The value of the parameter, 0.95 is from section 8.3 of methodology/5/. The section states that leakage shall be considered as default 0.95 in accordance with Section 5.4 of AMS-II.G. The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/ was determined in line with the applied methodology/5/.
8.	$f_{\text{NRB},y}$	0.8872	The parameter represents fraction of woody biomass that can be established as non-renewable biomass. The value is sourced from Third party report by C4 Eco solutions/27/ in line with TOOL 30, version 4.0/28/.The value is consistent in Joint PD-MR/1/ and ER calculation sheet/8/ was determined in line with the applied methodology/5/. The detailed assessment has been included below. According to the Third party report, the value is 91%; however, PP utilized the fNRB value from another registered project, which was less than 91%, as a conservative measure which was verified from the detailed excel sheet with list of registered projects and baseline fuel consumption value/43/. The value has been adjusted to 88.72% as a result.

Assessment of fNRB:

PP has submitted the f_{NRB} calculation sheet/44/ containing the sources of values, which was reviewed by the assessment team.

The formula used for calculating the fNRB was found to be in accordance with TOOL30, version 4.0/28/:

$$f_{NRB} = NRB / (NRB + RB)$$

Where,

f_{NRB} = Fraction of non-renewable biomass in the country/region or project area (fraction or %)

NRB = Quantity of non-renewable biomass (t/yr) in the country/region or project area.

RB = Quantity of renewable biomass in the country/region or project area.

Since NRB is determined on a country specific basis by the PP, the following method was applied in line with paragraph 9 of TOOL 30, version 4.0/28/.

$$NRB = H - RB$$

Where,

H = Total annual consumption of wood in the absence of the project activity in the country/region/project area (t/yr).

The NRB was calculated by the PP in-line to equation 2 of the applied Tool 30/28/ and was found to be acceptable as the f_{NRB} here is calculated for a specific country (i.e., Rwanda).

The consumption of woody biomass (H) was determined in accordance with paragraph 14, section 3.1 of TOOL30/28/, which provides two options to calculate this parameter. The parameter was determined using option (b) i.e., official statistics and reports. The calculations were checked from the calculation sheet/44/ and sources provided in that sheet as well as cross checked against online resources wherever deemed necessary by the assessment team.

The total annual consumption of wood in the absence of project activity in the host country was calculated based on total domestic fuelwood consumption by rural and urban population. The values for all these variables were obtained from World bank data/49/ and FAO/45/.

According to para 10 of the tool 30/28/, the most recent historical data available for the same vintage has been used for the f_{NRB} calculations i.e., values from year 2019. Two different sources for deriving the fuelwood consumption were used as national level data is available for consumption of fuelwood in Rwandan households.

The calculations for f_{NRB} value, as demonstrated in the calculation sheet/44/, have been done using the equations 1, 2, 3 and 4 of the applied tool version 4.0/28/. All parameters' sources used for calculations of f_{NRB} are listed in appendix III of the validation report.

Estimation of renewable biomass (RB) is carried out in line with paragraph 19 of TOOL30/44/:

$$RB = \sum(MAI_{forest,i} \times (F_{forest,i} - P_{forest})) + \sum(MAI_{other,i} \times (F_{other,i} - P_{other}))$$

Where:

$MAI_{forest,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of forest areas (t/ha/yr)

$MAI_{other,i}$ = Mean Annual Increment of woody biomass growth per hectare in subcategory i of other wooded land areas (t/ha/yr)

$F_{forest,i}$ = Extent of forest in sub-category i (ha)

$F_{other,i}$ = Extent of other wooded land in sub-category i (ha)

P_{forest} = Extent of non-accessible area (e.g., protected area where extraction of wood is prohibited, geographically remote area) within forest areas (ha)

P_{other} = Extent of non-accessible area (e.g., protected area where extraction of wood is prohibited, geographically remote area) within other wooded land areas (ha)

i = Sub-category i of forest areas and other wooded land areas

The total renewable biomass in the host country was calculated to be 578,495 t/yr.

According to the list of default country-specific fNRB values approved by the Board (table 1.0), Rwanda has the fNRB of 98% (though it is expired). <https://cdm.unfccc.int/DNA/fNRB/index.html/>.

Hence, the value of fNRB considered is consistent with the fNRB value of Rwanda from other sources.

Moreover, the fNRB value as per ASB0041-2018 has been calculated using version 1 of Tool 30. Also, the commercial wood consumption value which has been used for calculating 77% as per PSB0045 is based on rough estimations as stated in “Projection scenario of supply/demand of woody biomass In Rwanda from 2015 to 2026” (page 8). (CDM: Standardized baselines (unfccc.int))

fNRB calculation:

$$NRB = H - RB$$

$$= 6,177,233 - 578,495 \text{ t/yr}$$

$$= 5,598,738 \text{ t/y}$$

$$f_{NRB} = NRB / (NRB + RB)$$

$$= 5,598,738 / (5,598,738 + 578,495)$$

$$= 0.91$$

The calculated f_{NRB} of 0.91 indicates that the consumption of woody biomass within the country is greater than its capacity to supply sustainably. The third party f_{NRB} report/27/ has detailed the calculation which was found appropriate by the assessment team.

As per para 13 of the CDM tool 30, the NRB was compared to the top-down product of the most current annual deforestation data (3,187 ha/yr) and the average above-ground biomass (146.06 t/ha), in accordance with paragraph 13 of TOOL 30 v4/28/. The NRB, as determined by the CDM Tool 30 v4.0 2022 and reported in this report, is 5,598,738 t/yr, higher than the cross-check results based on deforestation (465,493 t/yr). Since the cross-check estimate of biomass loss from deforestation only takes into account the average biomass stocks from fully deforested areas, it is likely to underestimate the amount of non-renewable biomass. The cross-check does not take into account unsustainable wood harvesting that leads to forest degradation and significant drops in biomass stocks but does not completely eliminate deforestation. Consequently, as the instructions in CDM Tool 30 v4.0/28/ were followed, the estimate of NRB is deemed appropriate for this investigation.

A summary of sources and assessment of all parameters in line with applied TOOL 30/28/ mentioned above used in calculation of $f_{NRB,y}$ (for Rwanda) have been provided in table below:

Parameter and Description	Value	Data Source	Means of assessment
$f_{NRB,y}$ (Fraction of non-renewable biomass in the country/region or project area)	0.91	Calculated in line with TOOL 30	This parameter was found to be in line with equation (1) of TOOL 30/28/ as checked against f_{NRB} calculation sheet/44/.
NRB (Quantity of non-renewable biomass (t/yr) in the country/region or project area)	5,598,738	Calculated in line with TOOL 30	This parameter was found to be in line with equation (2) of TOOL 30/28/ as checked against f_{NRB} calculation sheet/44/.
RB (Quantity of renewable biomass in the country/region or project area)	578,495	Calculated in line with TOOL 30	The calculation of this parameter was found to be in line with equation (4) of TOOL 30/28/ as checked against f_{NRB} calculation sheet/44/. The weighted mean annual increment (MAI) values were calculated based on the MAI values taken from FAO document/45/, and forest extent

			covered under different ecological zone, the values of which were taken from 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories/48/.
H (Total annual consumption of wood in the absence of the project activity in the country/region/project area) (ton/yr)	6,177,233	Calculated	In line with TOOL30 para 14 under section 3.1 /28/, the value was calculated based on the sum of total domestic and non-domestic fuelwood consumption by rural and urban population, using the wood to charcoal conversion factor. The calculation of this parameter was found to be in line with equations of TOOL30/28/ as checked against fNRB calculation sheet/44/ and JPD-MR/1/.

All source links were assessed and discussed in detail against each input parameter in detail. These calculations were assessed and accepted by the assessment team. The total annual consumption of wood in the absence of project activity was calculated to be 6,177,233 t/yr as verified from fNRB calculation sheet/44/.

As per para 6 of the tool 30, comparison between peer reviewed literature (Bailis et.al.) has also been included in the PD. The fNRB value for the Rwanda is higher than the expected fNRB values determined Bailis et.al report using the WISDOM method. While direct comparisons between the WISDOM and CDM methodologies are only sometimes appropriate –given the different approaches of the methodologies–the following factors can contribute to variations in the fNRB estimates between the application of the CDM method in the present study and the WISDOM method:

- **Parameter H (Total consumption of woody biomass in the applicable area in the relevant period (tonnes)):** More recent population statistics (2021) for Rwanda were used and represents an increase in population numbers since the 2009 data was utilized in the comparison study. As the value of parameter H is determined using the entire population, it is therefore greater than the value obtained from the Bailis report.
- Updated 2019 FAO forest products statistics were used in the present study, whereas the comparison study used 2013 FAO forest products statistics.

- **Parameter RB (renewable biomass):** The present analysis gives lower estimates of the accessible forest area than the comparison study due to the methodology used. As a result, the predicted RB is reduced, which raises the NRB and fNRB.
- **Parameter MAI (Mean Annual Increment):** A weighted average based on the forest area of three categories—primary forests, secondary forests older than 20 years, and secondary forests younger than 20 years—was used to compute MAIs in the current study. A 21-year future period was projected using the observed forest increase extents between 2000 and 2012 using data from the 2019 Refinement of the IPCC Guidelines. The comparison study states that the assessment of growth rates as a percentage of standing stock was done differently after field measurements and IPCC numbers were combined to determine MAI values. This method frequently results in higher MAIs, which can then raise RB estimations and, as a result, lower NRB and fNRB predictions.

Since the project involves fNRB calculations based on national level data, VVB has cross checked the data with Global Forest Resources Assessment 2020: Main Report (FAO, 2020)/45/, Global Forest Watch, 2023/46/ and Mongabay, 2020/46/.

Furthermore, the values of fNRB were also compared against other registered projects across VERRA and Gold Standard in Rwanda /43/ and found that the values used in other registered projects across various standards range from 88.72% to 98%. In order to be conservative, the fNRB value has been considered as 88.72% which is the lowest value among all the values of fNRB amongst the registered projects and found it to be appropriate.

Data and Parameters to be monitored:

S.No.	Parameter	Monitoring frequency	Means of assessment
1.	$N_{y,i,j}$	At least once every two years	The parameter will be recorded in the database and updated for operational rate through monitoring surveys in-line with the applied methodology/5/.
2.	η_{old}	Fixed for each individual household at first HH visit	The value applied for the ex-ante estimates is the same as the actual value obtained at the time of distribution of ICS units for PAI1. However, the value of the parameter will be recorded based on the actual data obtained at the time of distribution for each PAI.
3.	$\eta_{new,i,j}$	Annually	The parameter represents the efficiency of the device of each type i and batch j. The efficiency of the distributed cookstove will be monitored as per as the equation 5 of the applied methodology/5/, which provides a

			provision of default loss in efficiency for each year. Design efficiencies in-line with the manufacturer's specifications/15/ have been used to arrive at the ex-ante efficiency value for each year within the crediting period /8/.
4.	μ_y	At least once every two years	This parameter represents the adjustment to account for any continued use of pre-project devices during the year y. The data will be monitored at least once every two years.

Baseline Emissions and Project emissions:

It was checked with the applied methodology that there are no baseline emissions and are not quantified separately. The emission reductions saved under the project are calculated through one equation which directly evaluates the GHG saving of amount of fuel saved on switching to new ICS as compared to the previous stove.

Leakage emissions:

Leakage shall be considered as default 0.95 in accordance with section 8.3 of VMR0006/5/.

The equation stated under section 5.4 of the Joint PD and MR/1/ were compared with the applied methodology/5/ and were found to be exactly same. The same equations have been used in the estimated ER sheet/8/ to calculate the estimated ERs for the entire crediting period.

3.4.7 Methodology Deviations

The VMR0006, v.1.1/5/ was adopted for the Group project activity, and the PP adhered to the VCS PDMR/1/ development process. The validation team has confirmed that the Group project activity has not been subjected to any methodology deviations.

3.4.8 Monitoring Plan

The monitoring plan covers:

1. Maintenance of database:

The database shall include the following:

- Unique project identification number
- Installation date
- Family name
- Contact details of user (when available)

- Socioeconomic level (if households or communities are targeted), or type of establishment (if SMEs are targeted)
 - Baseline cooking fuel source
 - Baseline stove type
 - Date of replacement of cook stove
 - Monitored parameters
 - Usage of Skirt with the project ICS
2. The roles and responsibilities of the parties involved is described in line with the VCS standard.
 3. The PP has various quality control measures in place like training staff to conduct surveys. Trained supervisors conduct spot check on surveyor results, all information related to parameters are collected on phone-based application/forms, surveys results are tagged as valid or invalid based on the duration of surveys, data validation done by PP. Trainings records will be maintained,
 4. Calculation of ERs is done on the basis of latest results by PP's team.
 5. Joint PD & MR/1/ has been reviewed to check that the procedure for data uncertainty, emergency preparedness, roles and responsibility, operational and management structure are mentioned in the Joint PD & MR. The monitoring plan completely describes all measures to be implemented for monitoring all parameters required.
 6. Sampling Plan:

Sampling plan is made according to the CDM Standard 'Sampling and surveys for CDM project activities and programmes of activities'/7/, and CDM Guidelines: Sampling and Survey for CDM project activities and programmes of activities/6/ to estimate monitored parameter values used in the ER calculations.

The target population will be the end users listed in the database on which random sampling will be applied.

The sample size equations stated in the Joint PD and MR/1/ were checked and found consistent with CDM Guidelines: Sampling and Survey for CDM project activities and programmes of activities, version 4/6/.

The sampling frames may include individual PAI or group of PAIs. The PP will apply 95/10 confidence/precision basis on a biennial basis. If sampling is done for groups of PAIs, then 95/10 will be applied in line with CDM Standard Sampling and surveys for CDM project activities and programmes of activities, version 9/7/.

The monitoring plan included in the section 6.3 of the Joint PD & MR /1/ is based on the VCS methodology /5/ which has been applied to the proposed VCS grouped project activity.

The monitoring arrangements described in the monitoring plan are feasible within the project design.

3.5 Non-Permanence Risk Analysis

This is not applicable for the proposed VCS project as it is Non AFOLU project.

4 VERIFICATION FINDINGS

4.1 Accuracy of GHG Emission Reduction and Removal Calculations

A. Data and parameters available at the time of validation are same as listed above under section 3.4.6. of this report

B. Data/parameter monitored for PAI 1 under MP1:

S.No.	Parameter	Monitoring frequency	VVB assessment						
1.	$N_{y,i,j}$	At least once every two years	The parameter represents number of project devices of type i and batch j operating during year y. The data is sourced from database/8/ and monitoring surveys/11/. <table><tr><td></td><td>Stoves distributed</td><td>Operational rate</td></tr><tr><td>PAI-1</td><td>130,633</td><td>100%</td></tr></table> The total number of stoves distributed under the current monitoring period for PAI 1 is 130,633/8/. It was found that 100% of the stoves were operational as checked from the monitoring survey results /11/ carried by the PP for the first monitoring period. There was no drop-off observed during the current MP. The monitored samples checked by VVB during on-site visit/17/ were also questioned on the operability of the project stove, and all the responses were checked with the available information in the monitoring surveys/11/ which was found as sufficiently following the requirements.		Stoves distributed	Operational rate	PAI-1	130,633	100%
	Stoves distributed	Operational rate							
PAI-1	130,633	100%							
2.	η_{old}	Fixed for each individual household at first HH visit	The parameter represents efficiency of baseline stove. The value of the parameter considered is 0.1 determined from baseline efficiency sheet/29/ recorded for each household at the time of project ICS						

			distribution inline to the applied methodology /5/. The value is consistently applied and reported between ER sheet/8/ and Joint PD and MR/1/. The monitoring samples checked by VVB at the time of on-site visit were questioned on the baseline device and baseline fuel type and all the responses were checked with the available information in the monitoring survey/11/. This was checked on-site/17/ and found as sufficiently following the requirements.								
3.	$\eta_{new,i,j}$	Annually	The parameter represents the efficiency of the device of each type i and batch j. The value of the parameter during the current monitoring period: <table><tr><th rowspan="2">Crediting Period Year</th><th colspan="2">Kuniokoa Efficiency value</th></tr><tr><th>With pot skirt</th><th>Without pot skirt</th></tr><tr><td>Year-1</td><td>0.4822</td><td>0.4080</td></tr></table> Kuniokoa ICS were only distributed under the PAI1. Hence, $\eta_{new,i,j}$ value is determined in line with equation 5 of the applied methodology for year 1 for Kuniokoa ICS model/5/.	Crediting Period Year	Kuniokoa Efficiency value		With pot skirt	Without pot skirt	Year-1	0.4822	0.4080
Crediting Period Year	Kuniokoa Efficiency value										
	With pot skirt	Without pot skirt									
Year-1	0.4822	0.4080									
4.	μ_y	Atleast once every two years	This parameter represents the adjustment to account for any continued use of pre-project devices during the year y. The value used for PAI-1 for the current MP is 0.0629. The values have been sourced from the monitoring surveys/11/. As checked from the monitoring survey results/11/ carried out for the first monitoring period, the end users were using the pre-project devices at around 6.3% of the total cooking events. The value of the parameter was found as acceptable, and sufficiently								

			fulfilling the methodological/5/ requirements.
5.	Date of commissioning of batch j	04-November-2022 (distribution of first ICS)	The latest date of commissioning of a device within the batch is used as the date of commissioning for the entire batch. The value of the parameter is 04/11/2022, as confirmed from distribution database/8/.
6.	Life span	Burn Kuniokoa – 10 years	As confirmed from the manufacturer specifications/10/, the lifespan of the Burn Kuniokoa Generation 2 is 10 years.

In conclusion, VVB reviewed all the evidence submitted by PP related to GHG emission reduction calculations and confirmed that all the parameters are correctly applied. Default values used in the calculation were identified correctly. The emission reduction calculation has been done in line with the applied methodology /5/.

Monitoring Plan for PAI 1 for current MP:

By the end of current monitoring period, the project activity has installed 130,633 number of the ICS / PAIs. As defined in the PD, the PP has applied sampling approach for measurement and determination of the parameter $N_{y,i,j}$ (i.e. Number of ICS installed and operating till the end of the current monitoring period - “Number of project devices of type i and batch j operational during year y”) and μ_y (i.e. Adjustment to account for any continued use of pre-project devices during the year y). PP has calculated the sample size considering the confidence level of 95% and precision 10% with simple random sampling technique using the CDM standard requirements/6,7/. As per sample calculator, sample size for $N_{y,i,j}$ came out to 96 and μ_y came out to be 7. Considering PP conducted common survey for both the monitoring parameter, total 120 samples were randomly selected /25/ from the distribution database. The monitoring surveys were conducted between 19/12/2022 to 23/12/2022.

The monitoring plan included under Appendix 1 of the Joint PD & MR /1/ is based on the VCS methodology/5/ which has been applied to the proposed VCS project activity.

The assessment team has checked the sample calculation as per the Guideline: “Sampling and Survey for CDM project activities and programme of activities” version 04/6/ and found out that the calculation of confidence/precision is consistent as per the monitoring plan.

4.2 Quality of Evidence to Determine GHG Emission Reductions and Removals

The assessment team confirms that the calculation and data is authentic. The quality of supporting documents submitted for validation and verification are adequate. The assessment team has checked the quality and maintenance of the supporting documents during the on-site visit to confirm the

authenticity of the documents and to check the appropriate calculations. The assessment team confirms that the proper evidence is available for the whole monitoring period and the same is verifiable and the data collection system meets the requirement of the monitoring plan and the applied methodology according to the assessment carried out.

The assessment team confirms the quality of evidence to determine the GHG reductions are satisfactory and the detailed information regarding the roles and responsibilities have been provided in the Joint PD-MR/1/.

5 VALIDATION AND VERIFICATION OPINION

Earthood Services Private Limited, contracted by DelAgua Trans Africa Stoves Pte. Ltd. (DATAS) (PP), has performed the independent joint validation & verification of the project activity “DelAgua Improved Cookstove Grouped Project”. The PP is responsible for the information and disclosures made to substantiate about the implementation of the project activity.

ESPL commenced the joint validation & verification based on the baseline and monitoring methodology VMR0006, version 1.1/5/ and VCS-PD & MR and ERs estimation and calculation spreadsheet/8/.

ESPL's validation approach is based on the understanding of the risks associated with reporting the project activity, estimates of GHG emission data and the controls to be implemented to mitigate these. ESPL planned and performed the validation and verification by obtaining evidence, physical site visit, other information, and explanations that ESPL considered necessary to give reasonable assurance that the estimated GHG emission reductions are fairly to be achieved.

Further, the validation and verification of the GHG statement was conducted in accordance with ISO 14064-3; 2019 by the ESPL assessment team.

Therefore, the VVB confirms that:

- the project activity is in accordance with all relevant host country criteria (Rwanda) and VCS rules and requirements.
- the project activity is in accordance with all conditions of the latest version of applied methodology.
- the environmental assessment is appropriate and sufficient.
- the monitoring plan is transparent and adequate.
- all information has been consistently applied in the VCS Joint PD-MR.
- the implementation of the project has been done as per description in the VCS Joint PD-MR.

Quantity of GHG emission reductions or removals in tCO₂ equivalents are estimated to be 17,192,130 throughout the crediting period and 2,149,016 is the annual average reduced ERs. The validated ex-ante emissions reductions are as follows:

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
04-November-2022 – 31-December-2022	117,023
01-January-2023 – 31-December-2023	2,136,377
01-January-2024 – 31-December-2024	2,708,533
01-January-2025 – 31-December-2025	2,595,794
01-January-2026 – 31-December-2026	2,555,901
01-January-2027 – 31-December-2027	2,542,201
01-January-2028 – 31-December-2028	2,508,991
01-January-2029 – 03-November-2029	2,027,310
Total estimated ERs	17,192,130
Total number of crediting years	7
Average annual ERs	2,149,016

Quantity of GHG emission reductions or removals in tCO₂ equivalent achieved by the project during the verification period is as follows:

Verification period: from 04/11/2022 to 31/12/2022

Verified GHG emission reductions and removals in the above verification period is reported below:

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2022 (04-November-2022 –	59,473	-	-	59,473

31-December-2022)				
Total	59,473	-	-	59,473

Comparison between the estimated ex-ante GHG emission reductions and removals and the achieved emission reductions and removals for the current monitoring period is reported below:

<u>Ex-ante emissions reductions /removals</u>	<u>Achieved emissions reductions /removals</u>	<u>Percent difference</u>	<u>Justification for the difference</u>
117,023	59,473	-49.18%	The emission reduction achieved during current monitoring period are ~49% lower than value estimated for same period is mainly due to difference in the operational days. During ex-ante estimation, emission was calculated considering 58 days whereas ex-post the average operational days is 28 considering stoves were distributed during the monitoring period.

Approved by



Dr. Kaviraj Singh

Managing Director

Earthood Services Private Limited

Date: 21-11-2024

Place: Gurugram, Haryana

APPENDIX I: LIST OF DOCUMENTS REFERRED

S.No.	Title of document	Version	Date
1.	VCS PD & MR	8.0	04/11/2024
2.	VCS Program guide	4.3	-
3.	VCS Standard	4.4	-
4.	VCS VVB Manual	3.2	-
5.	VMR0006 Methodology for Installation of High Efficiency Firewood Cookstoves	1.1	-
6.	CDM Guidelines Sampling and surveys of CDM project activities and programmes of activities	Version 4	
7.	CDM Standard Sampling and surveys for CDM project activities and programmes of activities	Version 9	
8.	Emission Reduction calculation sheet	Version 8	
9.	Date of first stove unit distribution	04/11/2022	
10.	Manufacturer's specification- - Burn Kuniokoa Generation 2 - SSM S32-X stove (https://www.ssmcook.com/product/wood-stove-ssm32-x/)	-	Last accessed on 14/02/2023
11.	Monitoring survey results	-	Version 1.0
12.	CHAPTER 6: FOREST AND PROTECTED AREAS https://www.rema.gov.rw/soe/chap6.pdf	-	Last accessed on 26/10/2023
13.	FRA 2020 Étude de bureau, Rwanda	-	Last accessed on 26/10/2023
14.	End user agreement – Carbon Waiver on the Distribution Poster agreement	-	-
15.	Manufacturer's agreement and Stove purchase invoices	-	-

16.	KML Files shared by PP for the districts where distribution has taken place under PAI-1	-	-
17.	VVB On-site visit evidences	-	-
18.	General Guidelines and Procedure for Environmental Impact Assessment, Republic of Rwanda	-	-
19.	PP Declaration on the project VCS 3699 not registered under any program	06/02/2023	-
20.	Project 3699 webpage – DelAgua Improved Cookstove Grouped Project: https://registry.verra.org/app/projectDetail/VCS/3699	-	Last accessed on 14/02/2023
21.	http://www.reg.rw/what-we-do/biomass/	-	Last accessed on 14/02/2023
22.	https://www.mininfra.gov.rw/digital-transformation-1-1	-	Last accessed on 14/02/2023
23.	https://www.statistics.gov.rw/publication/eicv-5-main-indicators-report-201617	-	Last accessed on 14/02/2023
24.	Baseline KPT survey sheet KPT calculation excel sheet	Version 1.0	-
25.	Random sample generator	-	-
26.	2006 IPCC Guidelines for National Greenhouse Gas Inventories; Volume 2 Energy	Chapter 1 Introduction Chapter 2 Stationary Combustion	Last accessed on 14/02/2023
27.	C4 Eco Solutions fNRB calculation report	-	07/06/2023
28.	Tool 30 Calculation of the fraction of non-renewable biomass	4.0	-
29.	Baseline stove efficiency	-	-

30.	VCS Exemption Mail and letter	16/02/2023	-
31.	LSC meeting invite evidences- Newspaper invite FM announcement	13/02/2023	-
32.	https://www.cleancookingalliance.org/binary-data/RESOURCE/file/000/000/36-1.pdf	-	Last accessed on 26/10/2023
33.	https://www.climateinvestmentfunds.org/sites/cif_enc/files/fip_final_rwanda.pdf	-	Last accessed on 26/10/2023
34.	Gakenke LSC module, Gicumbi LSC module and Musanze LSC module	26/07/2022 27/07/2022 29/07/2022	-
35.	Local Stakeholder Meeting photographs and videos	Various	-
36.	Attendance sheet of LSC Meeting	Various	-
37.	CREEC efficiency test report of BURN Kuniokoa G2 stove with pot skirt	-	25/05/2018
38.	KIRDI efficiency test report of BURN Kuniokoa G2 stove without pot skirt	-	04/09/2019
39.	CREEC efficiency test report of SSM S32-X without pot skirt	-	14/04/2022
40.	National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda	-	December 2020
41.	Improved Cookstove Grouped Project In Rwanda (VCS ID: 3654) https://registry.terra.org/app/projectDetail/VCS/3654	-	Last accessed on 05/02/2024
42.	Installation Of High Efficiency Wood Burning Cookstoves In Rwanda (VCS ID: 2380) https://registry.terra.org/app/projectDetail/VCS/2380	-	Last accessed on 05/02/2024
43.	Summary_Baseline Fuel Consumption_VCS+GS4GG	-	Last accessed on 05/02/2024
44.	Rwanda f_{NRB} Calculation sheet	-	-

45.	FAO Resource Assessment Report 2020 for Rwanda https://www.fao.org/3/ca9878fr/ca9878fr.pdf	-	-
46.	Mongabay 2020 Annual report	-	Dated: 2020
47.	KPT protocol: https://cleancooking.org/binary-data/DOCUMENT/file/000/000/604-1.pdf	-	Last accessed on 05/07/2024
48.	2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories	-	Last accessed on 05/02/2024
49.	World Bank Data	-	Last accessed on 05/02/2024
50.	a. https://meecs.org.uk/wp-content/uploads/2022/04/Cooking-Diary-Study-Rwanda.pdf b. National Survey on Cooking Fuel Energy and Technologies in Households, Commercial and Public Institutions in Rwanda , Ministry of Infrastructure and Ministry of Finance, December 2020	-	a. Last accessed 31/07/2024 b. Dec, 2020
51.	Training records (modules, certificates, photos) of PP staff members	-	Various
52.	Baseline KPT Survey sheets- system (application) generated	-	Various
53.	VVB Random sample generator	-	18/11/2024

APPENDIX II: ABBREVIATIONS

Abbreviations	Full texts
BE	Baseline Emission
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification Request
CO ₂	Carbon di oxide
CP	Crediting Period

DR	Desk Review
EB	Executive Board
EF	Emission Factor
ER	External Resource
EIA	Environmental Impact Assessment
ESPL	Earthood Services Private Limited
FAR	Forward Action Request
GHG	Green House Gas
GP	Grouped Project Activity
ICS	Improved Cook Stoves
MR	Monitoring Report
PAI	Project Activity Instance
PD	Project Description
PP	Project Proponent
tCO ₂ e	Tonnes of Carbon di oxide equivalent
TA	Technical Area
TR	Technical Reviewer
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Units
VVB	Validation and Verification Body

APPENDIX III: COMPETENCE OF TEAM MEMBERS AND TECHNICAL REVIEWERS

Competence Statement	
Name	Jahnabi Kalita
Education	M.Sc. Environment Management
Experience	1+ year
Field	Environment, Climate change
Approved Roles	
Team Leader	YES

Validator	YES		
Verifier	YES		
Methodology Expert	NO		
Local expert	YES (India)		
Financial Expert	NO		
Technical Reviewer	NO		
TA Expert (X.X)	NO		
Reviewed by	Shifali Guleria, Quality Manager	Date	06/05/2023
Approved by	Deepika Mahala, Technical Manager	Date	06/05/2023

Competence Statement	
Name	Vishnu S Panicker
Education	M.Sc (Sustainable Development and Environment Management) B.Sc (Forestry)
Experience	2.5+ years
Field	Forestry and Environment
Approved Roles	
Team Leader	Yes (VM)
Validator	Yes (VM)
Verifier	Yes (VM)
Local expert	Yes (India)
Financial Expert	NO
Technical Reviewer	YES
TA Expert (X.X)	Yes (VM TA 1.2, 3.1)

Reviewed by	Shifali Guleria, Quality Manager	Date	19/04/2024
Approved by	Deepika Mahala, Technical Manager	Date	19/04/2024

Competence Statement			
Name	Deepika Mahala		
Country	India		
Education	M. Sc. (Environment Management), GGSIP University B.Sc. Hons. (Chemistry), Sri Venkateshwar College, DU		
Experience	6 Years +		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	ACM0002, AMS.I.D., AMS.I.A, AMS.III.AV, AMS.II.G, AMS-II.C		
Local expert	YES (India, Bangladesh)		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (TA 1.2 & TA 3.1)		
Reviewed by	Shifali Guleria (QM)	Date	28/04/2022
Approved by	Kaviraj Singh (MD)	Date	28/04/2022

Competence Statement	
Name	Olivier Uwimana
Education	Bachelor's in Applied Mathematics
Experience	4+ years

Field	Statistics		
Approved Roles			
Team Leader	No		
Validator	No		
Verifier	No		
Methodology Expert	No		
Local expert	Yes (Rwanda)		
Financial Expert	No		
Technical Reviewer	No		
TA Expert	No		
Reviewed by	Shreya Garg	Date	19/08/2019
Approved by	Anshika Gupta	Date	19/08/2019

Competence Statement			
Name	Shifali Guleria		
Education	M.Sc. (Environmental Studies and Resource Management), TERI University		
Experience	3+ year		
Field	Climate Change		
Approved Roles			
Team Leader	YES		
Validator	YES		
Verifier	YES		
Methodology Expert	YES (AMS-I.A., AMS-II.G., AMS-II.E., AMS-III.A.V., AMS-I.D, ACM0002)		
Local expert	YES		
Financial Expert	NO		
Technical Reviewer	YES		
TA Expert	YES (1.2, 3.1)		
Reviewed by	Deepika Mahala	Date	16/02/2022
Approved by	Ashok Gautam	Date	18/02/2022

APPENDIX IV: LIST OF FINDINGS

CAR: Corrective Action Request

CL: Clarification Request

FAR: Forward Action Request

Table 1. Remaining FAR from validation and/or previous verification

FAR ID	NA	Section no.	NA	Date : DD/MM/YYYY
Description of FAR				
NA				
Project participant response				Date : DD/MM/YYYY
NA				
Documentation provided by project participant				
NA				
VVB assessment				Date: DD/MM/YYYY
NA				

Table 2. CL from this verification

CL ID	01	Section no.	3.3.2	Date : 08/05.2023
Description of CL				
1. Para 3.18.3 of VCS Standard, v 4.4 “The project proponent shall conduct a local stakeholder consultation prior to validation as a way to inform the design of the project and maximize participation from stakeholders. Such consultations allow stakeholders to evaluate impacts, raise concerns about potential negative impacts, and provide input on the project design.” Please explain the compliance with standard why LSC meeting (held on 01/03/2023) was not conducted prior to the start date for the proposed Grouped project activity (04/11/2022). 2. The attendance list shared as evidence for the local stakeholder meeting A displays that the LSC took place on 26,27 and 29th July, 2022 for PA with VCS ID 2749. It also states that the distribution will begin from 28/08/2022. While the start date of the proposed PA (VCS 3699) is 04/11/2022. Please clarify if LSC meeting with end users was conducted for this proposed PA (VCS 3699). Please demonstrate with appropriate evidences.				
Project participant response				Date : 10/05/2023

1. The PP conducted Local Stakeholder Consultation meetings at two different levels:

A. At district level, a month before the distribution in the concerned district.

B. At national level, once at the time of submission of 'Request for Registration and review of Verification' by Verra

These two categories of meetings were held for the implementation of the VCS grouped project activity 3699. Category A meetings were multiple in numbers, subjected to each district where the distribution was being held for the first time by the PP under this project in the host country Rwanda.

The national level meeting was aimed at a centralised group of stakeholders, namely the government stakeholders, the NGOs, academia, the media, the developers of other GHG mitigation programs in the host country Rwanda, and any other entity which deems itself as relevant to the implementation of the VCS grouped project activity. The structure of meetings was segregated in two types as a means to include more and more stakeholders to be heard at the local levels, rather than just going ahead with the stakeholder meeting at the national level, where only some of the stakeholders have a rather limited access to attend the meeting.

Further information on the above is provided in the revised PD&MR. Kindly refer to that to have a reasonable level of assurance on PP hosting the LSC meetings for the VCS grouped project 3699.

The LSC meetings at the district level were conducted prior to the start date of the VCS GP implementation. This can be furnished with the help of the evidences made available to the VVB to establish the occurrence of the LSC meetings prior to the proposed/actual start date.

The national level meeting could not be completed prior to the start date of the VCS GP because of planning related reasons, mainly the delay from Verra's end on the verdict on VCS 2749 Request for Registration and review of Verification', as this GP (VCS 3699) was initially planned to be a PAI or PAIs under that GP. This was also communicated to Verra through an exemption request, where Verra was requested to allow the VCS grouped project 3699 to be submitted for the request for registration and review of verification prior to the national level LSC meeting. Verra misunderstood the request, since DelAgua never stated that will not conduct the LSC meeting. DelAgua maintains that the LSC meetings were conducted prior to the start date of the VCS grouped project activity 3699, which is evidenced through the invite posters labelled at the district offices a month in prior to the meeting date.

Verra's rejection of exemption request on conducting LSC post submission of "request for registration and review of verification" has not changed the factual information. DelAgua is now adding all the information and records available with regards to the LSC in the updated version of PD&MR, and this is now being shared with the VVB for their validation.

DelAgua will abide by the Verra advice on the revision of project documents to establish that the requirement on the LSC prior to the project start date or validation has been fulfilled. The Verra review for the project will resume upon reuploading the PD&MR and the FVVR with details on the Verra standard LSC requirements met by the PP.

2. The LSC meetings were conducted on the dates mentioned (i.e. 27, 26, & 29 July 2022). The information provided therein on the distribution was tentative, and was changed to meet the availability of the ICS units supplied by the manufacturer (Burn). The meetings with expected end users were conducted before the start date, and the evidence on the same are being shared with the VVB.

Documentation provided by project participant

NA

DOE assessment

Date: 15/05/2023

1. PP has conducted 2 local stakeholder meetings:

Meeting A at district levels on 26/07/2022, 27/07/2022 and 29/07/20022 for targeted end users

Meeting B on 01/03/2023 at country level for other relevant stakeholders.

The proposed PA was planned to be included as a PAI under the grouped project (VCS 2749) where the distribution of the project ICS was tentatively planned on 28/08/2022. Hence the agenda covered during Meeting A informed the targeted beneficiaries about distributing Improved Clean Cooking Stoves to the local people in Rwanda, for the VCS Grouped Project Activity 2749. However, the LSC meeting conducted on 26/07/2022, 27/07/2022 and 29/07/20022 at 3 districts deciphered sufficient information about the working of the portable ICS which is also distributed under the proposed PA (VCS 3699). The LSC meeting also collected feedback, queries, concerns from the end-users as confirmed from the LSC modules shared by the PP.

There were further planning related issues which were beyond PP's control that caused inherent delays and prevented the Project Proponent (PP/DelAgua) to comply with the prescribed VCS requirement of conducting local stakeholder consultation prior to the start date of the proposed PA (VCS 3699). PP has sought exemption from VERRA on Local Stakeholder Consultation requirements detailing the reasons for the noncompliance. Thereafter, PP conducted the county level LSC meeting on 01/03/2023 which was attended by various relevant stakeholders. Evidence including newspaper invites and FM announcement notifying the date, time and location of the event and photographs of the meeting B was shared by the PP.

2. It was confirmed from the photographs and attendance sheet shared by the PP that LSC was conducted on 26/07/2022, 27/07/2022 and 29/07/20022 for targeted end users.

Therefore, CL#01 is closed.

CL ID	02	Section no.	3.1	Date : 14/02/2023
Description of CL				
The distribution database shared doesn't record end-user name, phone, address/ phone number. PP is requested to justify the eligibility criteria 1(f) and 4 reported under section 1.4 of the joint PDMR for the proposed PAI.				
Project participant response				Date : 16/02/2023

The copy of distribution database includes only the critical markers, as it was used for Sample Selection.

The complete Household Visit database is being shared with VVB now.

All the information related to the end user (including any corrections from the government list) household is provided in the HHV database.

Following end-user related information is provided in the DB:

1. Photo of the government issued ID card of the beneficiary HH member
2. Province information
3. District information
4. Sector information
5. Cell information
6. Umudugudu information
7. Telephone number
8. Photo of the poster that was signed during the distribution, etc. are captured in the HHV records.

Documentation provided by project participant

NA

DOE assessment

Date: 20/02/2023

The distribution database shared by the PP was found to be complete and accurate with the information stated under section 1.4 of the joint PDMR.

CL#02 is thus closed

CL ID	03	Section no.	3.4.6	Date : 10/06/2024
Description of CL				

After reviewing the sales database, the VVB team observed that under column “CurrentBarcode” 413 unique IDs of distributed project stoves are duplicate entries. It is unclear, how PP is adhering to the eligibility criteria 1 (f) as per section 3.6.16 and 3.6.17 of the VCS Standard v 4.4 “To avoid double counting of emission reductions through the method of distribution, each cookstove has a unique identifier number and the distribution of each stove is logged onto a database which includes unique information for each recipient”. Few of duplicate stove IDs are as follows:

MQ000626

MQ000626

MQ000670

MQ000670

MQ001584

MQ001584

MQ001674

MQ001674

MQ002211

MQ002211

MQ002341

MQ002341

MQ002341

MQ002514

MQ002514

MQ002538

MQ002538

MQ002558

MQ002558

MQ003787

MQ003787

Project participant response

Date : 10/06/2024

There was a clerical error in the ER sheet which has now been corrected. Few data points were pasted twice causing the issue. The separate database which was shared with VVB had correct data points i.e. 130,659 households and had no duplicates, however, in ER sheet 213 HHs were pasted twice. PP has corrected the issue in the ER sheet and shared the revised documents.

Documentation provided by project participant

NA

VVB assessment

Date: 20/06/2024

The VVB now confirms after the review of the revised sales database, there are no duplicates stove IDs, thus adhering to the eligibility condition 1 (f) as per section 3.6.16 and 3.6.17 of the VCS Standard v 4.4. The PP has removed 213 duplicates of households from emission reduction calculations for the current MP and thus the ERs achieved has changed to 59,485 from 59,582 tCO₂e. The finding is thus closed.

Table 3.CAR from this verification

CAR ID	01	Section no.	-	Date : 31/01/2023
Description of CAR				
1. PP is requested to use the latest version (version 4.2) of joint PD and MR.				
Project participant response				Date : 09/02//2023
The latest version (4.2) of the VCS Joint PD & MR template is now applied to the PD & MR.				
Documentation provided by project participant				
NA				
DOE assessment				Date: 14/02/2023
PP has used the latest version, version 4.2 of joint PD and MR and filled the information in line with the template guidelines. Hence, CAR#01 is closed.				

Table 4.FAR from this verification

FAR ID	01	Section No.	3.1	Date : 05/11/2023
Description of FAR				
The next verifying VVB must check how already existing households with end-user agreements with DelAgua Health Rwanda (Voluntary) Limited were informed about the change in PP of the grouped project activity to DelAgua Trans Africa Stoves Pte. Ltd. (DATAS).				
Project participant response				Date : DD/MM/YYYY
NA				
Documentation provided by project participant				
NA				
DOE assessment				Date: DD/MM/YYYY
NA				